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Digital Humanities 2025: Book of Abstracts

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Conference Programme

Call for Papers and Motto

The call for papers was launched on the 20th of September 2024, with submissions accepted until the 8th of December 2024 (after an extended deadline). It was translated into English, French, German, Italian, Spanish and Portuguese thanks to the collaboration of several colleagues from the Program Committee and the Local Organizing Committee.

The motto "Building Access and Accessibility, Open Science to all Citizens" was inspired by the 2002 Budapest Open Access Declaration, but also by a commitment to make visible several questions regarding accessibility needs (physical and digital). The Declaration amplified the aim of making research more widely available and free to anyone with internet access. Lately, Open Science or Open Scholarship have reframed concepts such as openness, access and accessibility from a technical and ethical perspective, taking into consideration data, infrastructure, and/or collaboration. National, international, or organizational manifestos, statements, declarations, principles, and policies related to scholarly objects, practices or methods are being formulated to improve and accelerate research through increased transparency, collaboration, and a more inclusive access to scientific knowledge of our societies.

Keynote Speakers

As well recognized scholars within the Digital Humanities field, Javier Cha (Assistant Professor of Digital Humanities, Department of History, The University of Hong Kong) and Roopika Risam (Associate Professor of Film and Media Studies and of Comparative Literature, Digital Humanities and Social Engagement Cluster, Dartmouth College) were the two keynote speakers for DH2025.

In the opening session, **Javier Cha** explored the impact that transformer-based machine learning brings to the work of historians. Abstract provided by Cha: "As historians increasingly encounter vast amounts of digitized and born-digital

sources, the challenge has shifted to developing strategies for making sense of large, complex collections with the nuance that historical inquiry demands. The discussion begins with an earlier phase of my research, which aimed to engage in digitally mediated multiscale exploration ("digital (re)reading") through graph queries and data reuse. Using structured and relatively unambiguous sources, such as biographical data modelled in Neo4j, this phase underscored the potential of digital historical research to uncover latent structures and reveal surprising connections in a manner that preserves the historian's interpretive agency. Building on this foundation, I then turn to the present, where my team and I are focused on leveraging large language models (LLMs) and vision-language models (VLMs) to assist with "algorithmic reading" across heterogeneous and semantically complex corpora. This next phase of inquiry explores the affordances of LLMs and VLMs for conducting semantic, stylistic, sentiment, and multimodal analysis, moving decisively beyond the limitations of keyword-based search and frequentist approaches. Whereas the earlier digital macroscopes allowed users to zoom in and out of structured datasets, transformers enable engagement with more affectively and rhetorically rich sources, such as memorials, petitions, contracts, philosophical treatises, ritual guidelines, and poetry. Finally, I introduce the modular artificial intelligence (AI) framework developed in the DeepPast project, which promotes the use of pluggable, task-specific components running on low-power hardware rather than a hyperscale, monolithic, general-purpose system. The DeepPast architecture supports varying interpretive modes in a flexible environment where the historian purposefully engages in conversation with an AI assistant and research partner—one capable of offering critique, reframing questions, and proposing alternative perspectives. The lecture concludes with a set of guiding principles designed not only to keep the human in the loop but also to produce AI-assisted historical research marked by greater interpretive sophistication."

In the closing session, **Roopika Risam** led us through an overview about the institutions that we have long relied on to sustain knowledge, higher education and data, and that are now under attack. Abstract provided by Risam: "As digital humanists and citizens of the world, we have an important choice to make: do we keep reproducing the extractive and colonial systems in which we work, or should we build something else? Risam will argue for an approach to digital humanities that is grounded in justice, tying access to accountability, repair to care, and scholarship to solidarity. The choices we make about digital accessibility, inclusive platforms, and the role of equity and diversity in our work will determine which types of knowledge persist in a world unmade and whose voices and stories survive for the future."

DH2025
ACCESSIBILITY
& CITIZENSHIP

Table of Contents

Pre-Conference Workshops and Tutorials 1

Mini-conferences 44

SIG Workshops and Events 65

Panels 69

Long Presentations 115

Short Presentations 561

Posters 1197

What do you do with 8 thousand billion variants? Toward structural and quantitative philology

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The paper will present the first mounting of a new philological method tackling the issues of managing too many variant readings. The traditional philological method of stemmatology is based on transmission errors: an editor uses them as a guide to reconstruct lost originals (Trovato, 2014). These errors are normally found by collating word by word from the texts found in a given work's witnesses (manuscripts or print). These operations have recently been subjected to a methodological revamp thanks to the arrival of powerful tools for ATR, alignment and stemmatology; however, these prove unusable on their own when dealing with fluid traditions and large datasets, and new methods must be envisaged.

The case study that allows us to exemplify the problem and the possible solution is offered by *Il Capitolo dei Frati*, a poem by the Jesuit Sebastiano Chiesa, written after the 1650s in which he satirises the chapter of Franciscan friars, which took place in 1645 in Modena. The work was a best seller at the time: we have located over 250 manuscripts written between 1670 and 1854, but we suspect the existence of over one thousand copies. Practically, every monastery and convent of any religious order in Italy and beyond had at least one copy, if not more: we find copies in London, Paris, Madrid, Dresden, Philadelphia, Harvard and Toronto, for instance. We have directly examined 140 copies, and a collation by *loci critici* has demonstrated that we are dealing with a large variety of text types. To choose our *loci*, we have adopted the Andrist and Maniaci (2021) structural textual segmentation, which subdivides text into blocks of content (BlockCont) and chooses the liminal section of each of them as our loci. For our preliminary collation, we therefore considered the incipit, the incipit of the so-called *Argomento* (i.e. a stanza providing an abstract of each canto), as well as the explicit (last verse) of each canto. For each *locus*, we have determined a certain number of types that we call "canonical versions" and noted aside the smaller variants of the canonical form. For instance, there are two canonical versions of the incipit of the Canto Sesto: (1) "Già è scorso il tempo e già venuta è l'ora", and (2) "Chi chiama in suo favor Sant'Agostino", each of them may have local variations. The poem presents itself in a varying number of cantos, 10, 12, 14, 15 or 16, each of which presents different incipits and different lengths. The following table summarises all the canonical version types we have considered for each *locus*. We used the 16 canto versions (C1, C2, C3, etc.) as our reference.

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	Sum
Argomento	3	2	2	2	2	2	1	3	3	1	4	2	1	3	2	1	**34**
Incipit	1	1	1	1	1	2	1	2	1	1	1	1	1	2	1	1	**19**
Explicit	4	2	4	2	3	3	3	3	4	2	5	1	5	4	2	3	**50**
Combinations	**12**	**4**	**8**	**4**	**6**	**12**	**3**	**18**	**12**	**2**	**20**	**2**	**5**	**24**	**4**	**3**	

Figure . Canonical types for the Capitolo dei Frati

The last row presents all possible combinations of incipits and explicits for each canto. Once all the options are combined, we find that we can have up to 8,255,648,563,200 possible combinations. However, these combinations also need to be multiplied by the distinctive lengths for each Canto. Canto 1, for instance, can have 27 different lengths, ranging between 54 stanzas and 150, which means it can have between 432 and 1200 verses. If we multiply these lengths (27) by the possible combinations of incipit/explicit/arguments for C1 (12), we find that we have 324 possible versions only for Canto 1. Furthermore, these numbers still need to account for displacements and reordering of the ottavas, which occur very often. At present, of the 140 witnesses examined, we have found 136 different text types of C1, and only two witnesses are virtually identical for the full poem.

When a tradition is so fluid, collating the text and looking for mistakes or similarities is meaningless because the texts are so diverse on a substantial level that one cannot apply any of the collation or even quantitative approaches on a word or letter level (van Zundert 2020, Hyytiäinen 2023). To handle what we call *modern mouvance*, within the project PRIMA (https://prima.hypotheses.org/), we have elaborated a computational workflow that allows us to cluster versions and then choose the most representative one for each cluster. Since the concept of error is meaningless, we have decided to focus on similarity of features and then apply quantitative analysis to them, similar to what is now being tested for quantitative codicology (Maniaci 2021). We define features as observable and measurable facts about the text or codex. For

this experiment, we have considered textual, paratextual, and codicological features.

In the first category, for instance, we included the incipit and explicit of each canto and the number of stanzas each includes. In the paratextual category, we considered the title, running titles, indices, and the presence of texts like addresses to the readers. For the codicological features, we considered format, ruling methods, layout, etc. All this data has been structured and stored within a relational database. We transform each manuscript into a row of a structured table, where every Canto is represented by its features. The data are normalised in order to be recognised by the processed. We then mined the data of each separately, according to the three distinct categories, and noted the clusters that emerged. We use Scipy (https://scipy.org/) and Scikit learn (https://scikit-learn.org/stable/index.html) to handle the data and create a significant representation. The optimal number of clusters is defined by the distance between the separation of each cluster. First, the groups are represented with a dendrogram and a decision tree, which allows us to see on what features the algorithm is based. Then, the clusters are shown on a 3d plot, and we finally, for each cluster, take the manuscript which has the best meaning distance to the other, closest to the middle of the cluster. This “average manuscript” will be considered as the reference copy of each cluster. This operation is necessary to sort out at least some of the tradition's noise, allowing us to focus on a more manageable number of versions and readings.

Editing of the chosen versions will follow what has become the usual workflow: reference copies will be transcribed using ATR tools (eScriptorium: Stokes et al., 2021), aligned against each other using Passim (Smith, 2015-), and encoded in TEI before being published using TEIPublisher.

Within the PRIMA project, we will also test it against other works of the 17th and 18th centuries, characterised by the same high rate of internal variation and fluidity. The latter emerges as one defining characteristic of literary, popular texts transmitted by manuscripts of all periods, as the untreatable traditions of Medieval texts such as the Roman de la Rose or Fleur, Parsifal, Fleur et Blanchefleur suggest. Therefore, this workflow and this method may have wide methodological implications and potential for all traditions featuring such a high level of variants.

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The Unnatural Language of Poetic Meters, Or Why You Should Be Afraid of Counting Words

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Introduction

The influence of poetic form on natural language is well-known. It has been examined as the "linguistics of verse" within structuralist metrics (Gasparov / Skulatcheva 2005; Gasparov / Tarlinskaja 2008), following Roman Jakobson's integration of poetics into linguistics (Jakobson 2010 [1960]; Hanson 2012). Similarly, the recognition that the recurrence of formulas in sung epics is conditioned by metrical positions formed a core component of Oral-Formulaic Theory (Lord 1960), prompting modern investigations into the interface between meter and language (see Frog 2021 for a review). Studies reveal how line division, caesura placement, and recurring rhythmic and rhyming patterns alter the flow of language by imposing constraints that modify the usual calculations for prosody, morphology, and syntax, shaping language into (somewhat) arbitrary forms. In this sense, poetic form renders natural language un - natural—a transformation that is, after all, one of poetry's primary aims: to craft markedly unordinary speech.

What is less often discussed are the consequences that the poetic form has at the level of a corpus, particularly in the context of quantitative and computational analysis. Different meters, line lengths, and stanzaic forms impose recurring patterns on a language already dense with structured relationships. Under metrical constraints, everything becomes confounded: the accentual classes of words, the rhythmic preferences of various parts of speech, and the syntactic gravity within poetic lines. In this paper we show how quickly these regularities accumulate into strong statistical signals and become a major source of linguistic variation, which remains largely unaccounted for in computational literary studies. Specifically, we demonstrate through a supervised classification of Czech, German, and Russian poetic texts that metrical forms easily manifest in seemingly unrelated and surface-level feature distributions.

Consider, for example, metrical poetry in Czech. Czech has fixed initial word stress, and implementing various prosodic patterns relies on alternating words of different lengths and padding lines with unstressed monosyllables, typically function words. Because of the fixed initial stress, it is relatively straightforward to begin a trochaic line, where stress is expected on odd counted syllables. However, starting an iambic line requires unstressed, prosodically weak monosyllables to appear at the very beginning. The weakest monosyllabic words are typically function words: articles, conjunctions, and prepositions. Since Czech does not have articles, conjunctions become the weakest class of words stress-wise. Poetic lines frequently begin with coordinating conjunctions like 'and', as this provides an easy way to form an independent clause and establish a connection between lines. By our calculations, 65% of all occurrences of 'a' ('and') in Czech iambic pentameter fall on the first syllable of a line. Consequently, it is unsurprising that conjunction frequencies increase in iambic lines compared to trochaic ones (Červenka 2006: 34–35).

In a corpus, these dependencies accumulate to a terrifying degree. The frequency of the conjunction 'a' alone is sufficient to distinguish iambs from trochees, even in relatively small samples (Figure 1a). The effect cascades through other levels of linguistic abstraction, influencing the frequencies of parts of speech and the general distributions of word lengths. Quite remarkably, the proportions of monosyllabic and disyllabic words can easily differentiate samples from all four major metrical forms in Czech: iambs, trochees, dactyls, and amphibrachs (Figure 1b).

a. *Stacked dot plot showing distribution of Czech 'a' across samples;* ***b.*** *Scatter plot showing relationships between frequencies of monosyllabic and disyllabic words. 60 independent random samples of 100 lines for each meterwere used.*

The analysis we present here explores the intuition outlined above: the entanglement between metrical form and linguistic features that are not directly derived from prosody, yet are still shaped by the gravity of a metrical template. These range from simple function words, to sequences of morphological features and word lengths. We ask how well we can recognize metrical form with surface-level features and at what sample size the signal becomes strong enough to worry. What we find, however, leads us to a different question: is there anything at all that can escape the grasp of a meter?

Data & Methods

Data. We analyze three poetry corpora with extensive metrical annotations: Czech, German, and Russian, all part of the multilingual collection PoeTree (Plecháč et al. 2024)[1]. We filter the corpora to include poets born in the 19th century and the most common recognized meters, each with at least 5,000 lines. Metrical annotations at the line level include both the foot type (e.g., iamb, trochee) and the specific metrical form (e.g., tetrameter, pentameter). Table 1 provides a summary of the data. For details on metrical form see (Šeļa et al. 2022).

Table 1. Corpus summary after filtering.

	Czech	German	Russian
# lines	2M	538k	660k
# foot types	4	3	5
# metrical forms	16	8	17

Features. Drawing on the systematic linguistic effects discussed above, we construct four feature sets:

1. Most frequent words (MFW): The 100 most frequent words, roughly corresponding to function words.
2. Part of speech tags (POS). Morphological tagging was done with TreeTagger, despite PoeTree currently using UDpipe for uniform multilingual annotation. The reason for this is that prosodic annotation for three languages is not yet aligned with UDpipe output.
3. Word lengths in syllables (SYL): The number of syllables in each word.
4. POS + SYL combination: A fusion of POS tags and syllable counts into a single token. "Once upon a midnight dreary" becomes "ADV_1 ADP_2 DET_1 NOUN_2 ADJ_2".

Each feature set is represented twice: as unigrams and rolling bigrams to indirectly capture local syntactic connectivity. Bigram features with long tails (MFW and POS+SYL) are capped at the 200 most frequent features.

Classification. We design two main classification tasks:

1. Foot type recognition: A simpler task involving 3–5 classes across corpora.
2. Metrical form recognition: A more complex task involving 8–17 classes; now the classifier needs to also distinguish samples of the same meter, but of different line length, e.g. iambic pentameter from iambic tetrameter.

We use Random Forests (RF) as the classification algorithm due to its efficiency and robustness in handling feature vectors of varying nature and length. RF employs bootstrapping for both features and samples during training, providing an internal estimation of prediction error on unseen data ("out-of-bag" classification), which we use to report final accuracy.

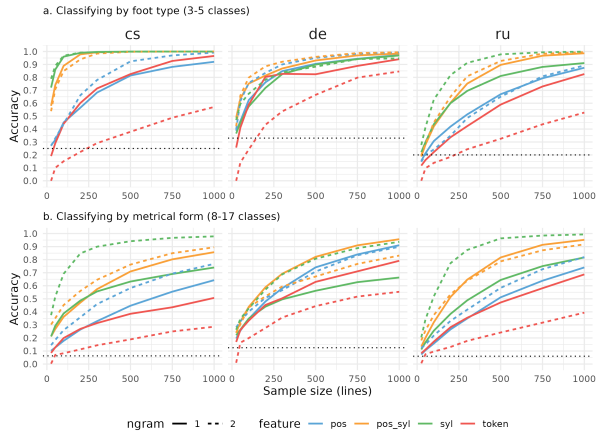
For each feature set, we take 5 independent samples of *N* random lines per meter/class, with *N* ranging from 25 to 1,000 lines. This reflects our filtering that ensured each class had a minimum of 5,000 lines. Each classification setup is repeated 100 times with new random samples, and average accuracies for each *N* are reported.

Results

The main result, illustrated in Figure 2, is clear: all features, even the weakest performers, reveal meter to some extent. With sufficiently large samples, many features achieve perfect or near-perfect predictions. Even in the challenging task of distinguishing between metrical forms performance remains surprisingly high. This finding confirms that the same prosodic pattern, when implemented in lines of different lengths, produces distinct morpho-syntactic affordances and linguistic profiles, since a line generally gravitates to form a complete clause, or a syntactic period.

As expected, word lengths in Czech—being the feature most closely tied to prosodic regulation—are the strongest predictor of foot type, demonstrating high efficiency even in small samples (~0.95 mean accuracy for N = 100). Generally, our

setups exceed random baselines even with the smallest, highly restrictive values of N, corresponding to sample sizes as small as single lyrical poems (Shelya / Sobchuk 2017).



Results of the RF classification. Horizontal dotted lines correspond to random baselines for each language.

In German and Russian, languages without fixed stress, sequential information (e.g., bigrams) becomes more critical, capturing patterns of phrasal stress and systematic relationships between morphology and stress placement (Gasparov / Skulatcheva 2005; Nenonkova et al. 2007; Haider 2021: 3721). While the performance curves for German and Russian do not appear as dramatic as those for Czech, this is a very strong performance: all but one feature in German reach > .90 accuracy in foot recognition, while Russian slightly outperforms Czech when it comes to classification by metrical forms.

Discussion

We have shown the consequences of simple repetition of a rule, a tendency, a pattern in a corpus. The exceptionally high classification accuracy tells us that all metrical forms organize language in their own ways, making form a major source of pervasive linguistic variation that influences even the simplest measurements we take. This result might be considered in optimistic light by those who are interested in automated recognition of meters, opening the way for the use of non-prosodic features that nonetheless reflect prosody. For literary and historical research, however, our results mean that form is a dangerous confound; not accounting for it will surely lead to measurement errors. How critical these errors can be, remains to be seen, it likely will differ across individual cases. However it is clear that we should adjust approaches that rely on corpus-wide calculations and direct comparisons of formally heterogenous poetry data (e.g. Kao / Jurafsky 2015, Walsh et al. 2024). Even if we are not looking at meters, they are staring at us.

Notes

1. Data and code are avaliable: https://github.com/perechen/meter-and-language

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Embracing absence in the digital humanities

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This paper poses a simple question: how can we improve and nuance quantitative analysis by treating absence not as a lack of information, but as a different type of data?

At this moment in time, it still the minority of records that are digitised, and – whether it is due to poor OCR, incomplete metadata, or simply a lack of digitised material – the datasets we used are often incomplete (Bishop 2018). Quality is therefore a key concern, and the adage "garbage in, garbage out" is an often-heard refrain at DH conferences. The phrase succinctly conveys that if a model is trained on datasets that are unrepresentative – whether that is in terms of gender, time period, race, language, or geographical origin – the analysis and output will reflect this bias (Bode 2018). As such, there is a significant body of research exploring just how representative commonly used corpora, such as Wikidata and Google Arts and Culture, actually are (Zhang / Terveen 2021; Kizhner et al. 2021).

The biases in these large datasets have begun to be addressed through initiatives such as WikiProject Women, while the innovative work by researchers such as Kestemont et al. and Ted Underwood, have attempted to model around incompleteness (White 2018; Underwood 2019; Kestemont et al. 2022). A renewed interest in the merits of smaller – and more representative – datasets catered towards answering a single aspect of inquiry has also emerged, inspite (or perhaps because of) the fervour surrounding foundation models (Piper 2017; Bender et al. 2021). But just how complete can a dataset be? As Katherine Bode notes, "there is no direct or absolute way to determine how much, or what parts, of the past we construct for analysis", and in defining the bounds of a project, researchers introduce absences in the form of necessary omissions (Bode 2020: 103). Borrowing from the growing school of thought that conceptualises biases not as a fault but as a feature of datasets, we explore how absence can be used to understand the priorities and biases of our contemporary practices (Underwood 2019; Impett / Offert 2024).

This paper will use a seemingly complete dataset as a case study to investigate the complexity of absence and resulting biases. Drawing on the work in postcolonial theory, intangible cultural heritage, and anthropology, we will discuss the phenomenology of absence – treating it not as a lack of information, but as a series of relations that reflect multiple intersecting processes (Fowles 2010; Mataga 2014; Edwards / Mead 2015). By adopting this interdisciplinary approach, we outline a reflective methodology that address the gap identified by Panagiotidou et al. (2022) for digital humanities research that tackles uncertainty, especially that which arises from conflicting data and the impact of human decision making.

The dataset we use, compiled between 2022 and 2024, builds on the Mapping Museum Project to create a list of over 4,000 active UK museum websites (Candlin et al. 2019). Combining the use of computer vision on digital collection screenshots with an analysis of the HTML code, we have investigated the hosting solutions, languages, and collections information available across the sites (Schmidt et al. 2020). By comparing this digital repository to current records of museum collections, we have identified which types of collections are absent from our digital landscape, and which have been rendered nearly unfindable by poor digital infrastructure (Ballatore / Candlin 2023). In this way, the dataset is able to show how material and cultural biases are compounded by digital technologies and research practices.

In crude terms, even the entirety of the surviving historic record is a poor dataset for understanding the past. Culture, in its broadest sense, is only partially and incompletely represented by material objects, the majority of which do not survive to the present day. Of those, only a small fraction make it into the collections of museums and fewer still are digitised or made available (Jaillant 2022). Yet, the search and recommendation algorithms that mediate our online experiences exacerbate these absences further and shape our perception of the resources available.

This paper unpicks these complex intersections to highlight the limitations of cultural collections data, while exploring the potential of these datasets to help understand the biases introduced by the priorities of today's practices. Within the context of continued digitisation and growing linked open data initiatives, we argue transparency around data quality and especially researchers' decision-making processes will become increasingly important our analysis and interpretation.

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One tree to Yule them all? Reflexions on intertextuality and text transmission

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Introduction

Text transmission is the process through which a text is disseminated and preserved dynamically through successive reproductions in one or several media (oral or written; manuscript, print or digital). It typically involves modifications, especially when the chain of transmission of a text moves substantially across time, space and cultural contexts, switches between media of transmission or types of reproduction (from paraphrase or citation to copy in full).

A typical case of text transmission is the dissemination of literary works in Antiquity and the Middle Ages. In this process, scribes generally use an existing material copy of a text to produce a new one, inducing innovations that can be intentional (e.g. translation, prosification, rewriting) or not (e.g. copy errors). Philologists use a genealogical tree (*stemma codicum*) to represent the relations between the surviving copies, and their shared ancestries based on shared innovations.

The relation between a model (exemplar) and its copy can be construed as a specific subtype of intertextuality, i.e. the interrelation between texts in general. Yet, the genealogies seldom represent the relations between all versions of a work, including translations and rewriting, or similar types of derivation, and they even more rarely include other types of intextextual relations (such as inspiration, paraphrase,…), even if there are counter-examples to be found (Aubreton 1968; Bédier 1902; Treutler 1880). Consequently, stemmata represent texts as independent entities, each having its own single root (the authorial manuscript or the latest common ancestor of the surviving copies). This is a strong difference with phylogenetics, in which all species are ultimately united in a single tree of life, going back to the last universal common ancestor of all life on earth (`luca`).

The underlying assumption of independence – a conscious simplification made mostly for practical reasons – is not without creating taxonomic difficulties. There is no universally accepted definition in philology as to when two versions, linked by intertextual relations, constitute only two redactions of the same text or two distinct texts. For instance, should the assonanced version of the Song of Roland (Segre 1971) be considered a different text than the rhymed version? Should the Middle Welsh translation (or even its descendants) be considered a witness to the French?

Theoretic as this question may seem, it has very concrete implications, for instance in terms of counting surviving (and estimating lost) copies of historical texts. Recent research has attempted to create a dynamic model of text transmission, using a stochastic birth and death model, and has shown results that aligned with existing data in terms of tree topology and of median age and surviving number of documents, allowing inferences on the number of lost works and documents, on a case study of medieval French chivalric narratives (Camps and Randon-Furling 2022). Created using established philological concepts, this model considered all root nodes (of texts) to be independent.

Even though correspondence between the model and historical data was shown to be close, we have noticed discrepancies regarding the distribution of witnesses (i.e. surviving copies) per text. Could a model that additionally includes the notion of ‘speciation’ yield results that give a better representation of historical data, confirming the importance of intertextual relations beyond typical copy? In this context, speciation may be understood as representing the creation of a new text, through derivation from an existing witness of the parent text, and could stand for cases such as translation, rewriting, prosification, or even the writing of prequels or sequels.

For the present study, we build upon the following definitions (Christensen and Camps 2025):

- **work** a story, e.g. Perceval and the Grail
- **text** an expression of the story, in a given language and form, for instance Chrétien de Troyes’ *Conte du graal*; Wolfram von Eschenbach’s *Parzival*; or the anonymous *Perceval en prose*
- **witness** a manifestation of the text, through a given copy, for instance, Guyot of Provins’ copy of Chrétien’s *Conte du graal*, preserved in a physical document.

Yet, even with these specific definitions, used in our research, the level of generality of the model we are going to use make it fit to describe various types of derivation and applicable to other taxonomies.

Materials and methods

Datasets

We compiled sets of texts and their witnesses from data published in the Forgotten Books research project (Kestemont et al. 2022) as well as the Jonas (Section romane 2024) and Handschriftencensus (Busch and Wolf 2024) databases. It is to be noted that, since those datasets preexisted, we do not control the criteria used to define and distinguish texts. Instead, each database rest upon cumulative philological scholarship in their respective domains.

Table 1. Number of texts and witnesses for the three data sources.

Source	Texts	Witnesses
Forgotten Books	776	3,587
Jonas	24,514	73,150
Handschriftencensus	6,201	36,893

Despite their dramatically disparate sizes, the different datasets demonstrate a similar imbalance in the number of surviving witnesses per text: most texts have less than five witnesses, a small minority have dozens of extant copies. At the same time, the probability of having a hundred witnesses is not negligible, see Figure 1. This suggest a heavy-tailed, Pareto-like behaviour for the distribution of witnesses per text, which is a pervasive property in cultural diffusion phenomena (Bentley, Hahn, and Shennan 2004; Newman 2005; Pinto, Lopes, and Machado 2012).

Figure 1. Distribution of witnesses per text for the three data sources (in log scale). The linear tendancy of the curves suggests a heavy-tail behavior, rather than an exponential one as the simple birth-death model produces. The artificial dataset (in yellow), generated with known parameters, is one of those used for validation (see Figure 3).

From birth-and-death processes to Yule's model

Several attempts have been made at modelling manuscript traditions by stochastic processes since the seminal work of Weitzman in the 1980s (Weitzman 1987; Cisne 2005; Camps and Randon-Furling 2022). These approaches, up to differences in corpora of interest and implementation, including time variations of the parameters of the dynamics, essentially consist in describing text transmission by a birth-and-death process accounting for the continuous creation of new manuscripts, modelled as independent agents, copied from their predecessors with probability # per individual manuscript and unit time, along with their destruction and loss happening with rate μ.

Although these models have been quite successful at reproducing several features of known textual traditions, they predict an exponentially decreasing distribution of the number of extant witnesses per text, incompatible with empirical data (see Figure 1).

Interestingly, the first historical attempt at modelling evolutionary dynamics in biology through stochastic processes (Yule 1925) also aimed at understanding the power law distribution of species per genus. In addition to the simple birth-and-death dynamics, Yule process introduces a *speciation event* with rate # by which a new agent –representing a species– is created, belonging to a new class –representing a genus. The large time distribution of surviving individuals per class is then known to follow the *Yule-Simon* distribution, with asymptotic power-law behaviour.

In our approach we model simultaneously the creation of independent traditions, appearing with rate # as a single living agent, and the internal process of speciation within traditions with rate #. This implies adding two additional parameters to the birth-and-death model, resulting in the parameter set #=(#, #, μ, #).

Table 2. Main hyperparameters and parameters of the model. The values for (T act, T inact, K, N0) are the one used for estimations. The intervals for (#, #, μ, #) define the initial search space for each parameter in our inference procedure.

	Definition	Values
T act	Duration of the active phase (n. time steps)	1000
T inact	Duration of the inactive phase (n. time steps)	1000
K	Tradition's maximum population size	500000
N0	Initial number of traditions	1
#	Average number of new traditions per time step (root birth rate)	[0, 1]
#	Probability of a witness being copied (node birth rate)	[0, 0.05]
μ	Probability of a witness being destroyed (node death rate)	[0, 0.01]
#	Probability of a new witness to manifest a new derived text (speciation rate)	[0, 0.01]

Figure 2. Example of a tree generated by the model. The tree stands for a single, but the nodes are distributed in different that arise through speciation events, and are shown here in different colours

Likelihood-free inference for Yule's parameter

The obtained process introduces significant complexity to the inference problem: in addition to the four parameter to infer, several additional hyperparameters must be provided for the simulation (see): a number of active (and inactive, when there is only deaths) time steps which defines the time scale in which we operate, and a theoretical maximum population K and an initial number of traditions N0 to control the generation of new individuals. The interaction between these parameters and hyperparameters produces complex tree structures through time, making the likelihood function analytically intractable.

Several frameworks exist for inference in that case. Among them, Approximate Bayesian Computation (ABC, Sisson, Fan, and Beaumont 2018) allows us to perform a complete Bayesian analysis without explicitly evaluating the likelihood. Recently, Simulation-Based Inference (SBI, Cranmer, Brehmer, and Louppe 2020) methods have emerged as powerful alternatives to traditional ABC approaches.

These techniques leverage neural networks to automatically learn the relationship between simulation outputs and model parameters, effectively constructing an approximation of the posterior distribution of the parameters (#, #, μ, #). Here, we use a multi-round version of one methode called Neural Posterior Estimation (NPE). The aim of running several rounds is to iteratively focus simulations on promising regions of the parameter space, refining the posterior approximation by concentrating computational resources where they matter most specifically for each dataset. Some results obtained after 5 rounds of NPE are presented in Figure 3 on an artificial dataset with known parameters, and in Table 3 and Figure 4 for our three corpora. While the estimation of the root birth rate # requires further refinement, we recovers the key correlations between # and μ identified in previously in (Camps and Randon-Furling 2022), while additionally highlighting distinct regions of interest in the full parameter space (# , #, #, μ) that characterise the different manuscript traditions.

Figure 3. Parameter estimates for our Yule model (#: new tradition rate, #: copy rate, #: speciation rate, μ: death rate) using artificial data. Diagonal panels show marginal posterior distributions for each parameter, while off-diagonal panels show joint distributions between parameter pairs. Black star markers indicate the true parameter values used to generate the synthetic dataset, and happens to be inside the inferred zone.

Figure 4. Parameters estimates for our Yule model across the three datasets (Handschriftencensus, Jonas, Forgotten Books). Diagonal panels show marginal posterior distributions for each parameter, while off-diagonal panels show joint distributions between parameter pairs. Point markers indicate the point estimation in Table 3, which is the median estimates for each dataset. The pairwise relationships between #, # and μ reveal strong parameter dependencies, highlighting distinct regions of interest that characterize each corpus's unique transmission dynamics.

Discussion

Table 3. Estimations of the four parameters of the Yule model along with their credible intervals at 95%. The values presented here can be visualized along the diagonal of Figure 4.

	Forgotten Books	Handschriftencensus	Jonas
#	7, 3.10#1 # [1, 8.10#1, 9, 7.10#1]	5, 5.10#1 # [5, 8.10#2, 9, 5.10#1]	4, 7.10#1 # [5, 9.10#2, 9, 7.10#1]
#	3, 7.10#3 # [2, 8.10#3, 4, 2.10>#3]	8, 0.10#3 #[5, 3.10#3, 1, 2.10>#2]	6, 6.10#3 # [5, 7.10#3, 8, 2.10#3]
μ	1, 6.10#4 # [1, 5.10#5, 8, 7.10#4]	1, 6.10#3 # [1, 7.10#4, 4, 1.10#3]	5, 6.10#4 # [1, 1.10#4, 1, 3.10#3]
#	5, 1.10#4 # [3, 5.10#4, 8, 2.10#4]	6, 1.10#4 # [4, 2.10#4, 9, 6.10#4]	2, 1.10#3 # [1, 8.10#3, 2, 8.10#3]

To fully validate the significance of intertextual relationships in manuscript transmission, our investigation must proceed along three complementary axes. First, we need to apply our inference framework across different textual corpora (see Table 1). Second, we must compare the performance of our Yule-based model against other candidates, particularly the simpler birth-death process, using formal model selection techniques. Finally, different algorithm choices should be tested to ensure the robustness of our conclusions. We plan to further investigate in these directions to better represent text transmission. If our current insights are confirmed, the use of Yule's model may constitute an advancement in the modelling of text transmission, beyond the simple birth-and-death processes already suggested, as a case of cultural drift. From this basis, it will be possible to see if further refinements are necessary, such as the introduction of a process of selection (modelling cultural preferences), or more generally, the use of the very large array of phylogenetic methods currently available for modelling species diversification (Morlon et al. 2024).

Supplementary material

Code for the model and for the ABC method will be available at the Github repository of the LostMa project, https://github.com/LostMa-ERC/BayesYule, and archived on Zenodo, https://doi.org/10.5281/zenodo.15350478.

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Playing the Past, Predicting the Future: Sortes Texts in Virtual Reality

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Philology is fundamentally the science of texts, yet texts are inseparable from the documents that carry them. In a way, scholarly editions start from the most “internal” perspective, that of a text *in* a document, to then, in some cases, surpass the limitations of the latter in a reconstructive effort aimed at restoring an idea of lost textuality as synthesis of multiple extant documents. In our project we approach a historical text literally from the *outside* , that is, from the rooms in which the document was likely to be used, to then zoom in via the document to the text itself and its translation.

The texts we focus on belong to the genre of the *sortes*. The *sortes* are divinatory texts with roots in Late Antiquity (Heiles 2018 and Heiles et al. 2021). They were used throughout the Middle Ages for predicting the future and answering questions related to the everyday life and struggles of the people. These texts often required the user to anticipate the consultation of the text by performing specific rituals, such as cleansing, fasting, praying or wearing specific clothing. Furthermore, it was necessary to use randomising mechanisms to navigate non-sequential reading paths in the codex, in order to disclose the predictions. The paths usually stretched from a question chosen by the reader at the beginning of the text to an answer which was given at the very end, after the user had visited a multiplicity of interactive stations. Given the ludic nature of this genre, the *sortes* were reinterpreted as social games in the course of the Middle Ages. Faced with the challenge of editing one such work, the Prenostica Socratis Basilei (PSB) in its mediaeval German versions, we soon recognised the limitations of the editorial effort.

We do aim at publishing a scholarly edition in accordance with the computational paradigm (Cugliana 2023, Cugliana et al. 2024a and Cugliana et al. 2024b). The modelling of the text, which is the basis for the creation of a digital scholarly edition, establishes a theory and an interpretation of the text (McGann 2001). But this represents a too limited view on the documents being edited: complementary to the model of the *text* expressed in the computational scholarly edition, we model a hypothetical, yet evidence based, process of *using* the document. We complement the textual view of the document with a performative view, in which the text in the document is seen as a script for performing certain underspecified activities with non-deterministic outcomes. Thus, what we are presenting in this paper is an addition to the edition project, a new opening which we regard as a bridge between the fields of philology, media studies and media archaeology.[1]

The performative modelling led to the creation of a VR application[2] which we designed to simulate the contextualised historical use of a *sortes* text.[3] We chose a manuscript which is currently being edited in the aforementioned PSB project, as it was already available both as a transcription and in translation (Schneider 1976 and 1978). We situated the codex in three virtual contexts: a monastic library, an astrologer's laboratory, and a tavern. These symbolise the places where *sortes* texts, in general (as specific information on the contexts of use of specific witnesses are very rare), were likely to be used in the Middle Ages (Heiles 2018), thus simulating specific settings for the performances. The three selected contexts represent, respectively, the spiritual, scientific and playful interpretations of the *sortes* . In this way we seek to recreate the multisensory experiences and ritualistic interactions that characterised mediaeval engagement with this genre. Users interact with an immersive, animated 3D model of the

codex, employing dice and tools within these virtual spaces. The simulation system was developed in Unreal Engine 5.3 with Meta Quest 3 head mounted displays as the devices used in the development.

The modelling work we carried out in praxis led to a deeper understanding of our object of research and to the formulation of new research questions,[4] which we will address in our presentation. Our first research question challenges the role of digital presentation of multimodal historical objects carrying texts: how can we make texts and documents come to life for the modern audience? Clearly, we need to extend the attention beyond the merely textual dimension and start looking at these documents in their materiality, at what *surrounds* them and at how they afford certain performative acts. To put it simply, you need to be in the same room where the book is in order to read it. And of course, you could not read it without first opening it![5] But, and we come to our next research question: what aspects make materiality feel real? We claim that interaction is key. Real pages have their weight and you cannot feel it until you slip your fingers between them and turn them. A simulation is not a reconstruction of the real experience, but it offers the users additional aspects of the experience. In this sense, simulation is also storytelling, a reimagining, a way of engaging with the past that underscores the vitality of texts and documents and brings them closer to modern audiences.

To conclude, if the goal of scholarly editions is to save written culture and memory from their perpetual condemnation to oblivion (Petrucci 2004) then scholarly editing as a discipline should continually adapt and innovate its "mnemotechniques"[6] (Assmann 2007) to align with the evolving ways modern audiences receive and consume content.[7] In the digital environment, this evolution often occurs at the intersection of various media.[8] Furthermore, recognising the *intrinsic* multimodality of text – where communication extends beyond words to include visual, auditory and interactive elements – represents a pivotal shift in how we represent and engage with textuality.

Notes

1. For the recently established field of media archaeology cf., among others, Parikka (2012) and Marx (2024).
2. A walkthrough-video presenting the main features of the prototype is available at https://www.youtube.com/watch?v=yu2uHmaOksE (accessed November 27, 2024).
3. Thus, the general model of the performance was translated into a simulation of specific performances.
4. This is an important aspect of modelling in the (digital) humanities, as well as beyond, cf. Ciula et.al. (2023: 71).
5. A humorous comment is made in "Medieval helpdesk", produced by the Norwegian Broadcasting Corporation (NRK) from 2001, cf. https://tv.nrk.no/serie/oeystein-og-meg/sesong/4/episode/FUHA16000101 (accessed 25.11.2024); with English subtitles: https://youtu.be/pQHX-SjgQvQ (accessed 25.11.2024).
6. That is, the methods and practices through which knowledge and collective memory is preserved and transmitted.
7. An example of mnemotechnical adaptation is the instagram project "Ich bin Sophie Scholl", by the German Südwestrundfunk (SWR) and Bayerischer Rundfunk (BR), described in the blog article of the "Bildungsstätte Anne Frank": https://www.bs-anne-frank.de/mediathek/blog/zum-instagram-projekt-ich-bin-sophie-scholl (accessed 25.11.2024).
8. All media are multimodal and intermedial (Elleström 2021: 4). On digital intermediality in the context of modelling, see chapter 4 of Ciula et al (2023).

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Algorithmic Edition

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Introduction

Digital editions are a cornerstone of the Digital Humanities, offering enriched resources that combine content with context (Sahle 2017). These editions serve as essential tools for research, incorporating structured annotations that support both traditional inquiry and computational analysis (Pagel et al. 2020). However, their reliance on visual platforms prioritizes accessibility over the algorithmic possibilities of digital texts (Buzzetti/Rehbein 2008). To address this, scholars propose a “text-as-network” paradigm, emphasizing the interconnectivity of data points to unlock their computational potential (Witt 2018: 219).

Annotations play a crucial role in this context, as they provide added layers of meaning to specific parts of a text, such as historical context, linguistic features, or thematic connections (Reiter 2020: 126). These annotations not only support human interpretation but are also key to preparing texts for computational analysis, transforming them into machine-readable datasets that facilitate deeper exploration (Pagel et al. 2020). Therefore, annotations are not merely passive metadata but active participants in scholarly discourse, connecting texts with insights derived from close, data-driven analysis (Pagel et al. 2020; Reiter 2020: 135). This function becomes even more pronounced when text fragments and their annotated context could be extracted, reassembled, and grouped around specific topics or themes.

In this paper, we explore how digital editions can serve as a foundation for what we call **algorithmic editions**. Central to this transition is the ability to extract, combine, and contextualize annotated text fragments, enabling their use for human and computational analysis. By leveraging technologies and concepts such as Labeled-Property Graphs (LPG), Applied Text As Graph (ATAG) and Explicitly Notated Citations (ENC), we demonstrate how these approaches make it possible to utilize text fragments, bridging the gap between traditional digital and algorithmic editions.

Towards an Algorithmic Edition

While the vision of algorithmic editions is promising, significant challenges arise when examining the current state of digital editions and annotation practices. These challenges stem from technical and yet unresolved limitations, which must be addressed to further explain the potential of an algorithmic edition.

Anchoring Problems

A foundational issue is the fragility of anchoring third-party annotations to textual fragments. While scholars conceptualize annotations as precise references to specific parts of a text, the reality of implementation is often tied to the technical framework of web browsers. As Koolen and Boot (2020) point out, browsers only interpret annotations as parts of a webpage's HTML structure, creating a disconnect between scholarly intent and technical execution. This reliance on HTML structure leads to instability, where changes in layout or relocation of web content render anchors invalid, undermining their reliability for long-term use. Moreover, one of the greatest challenges in this context lies in the flexible granularity when referencing these fragments, which cannot be adequately identified due to the lack of chapter, paragraph, or line references (Bernhart/Hahn 2014: 227; Ralle 2016: 155).

Lack of Persistent Linking

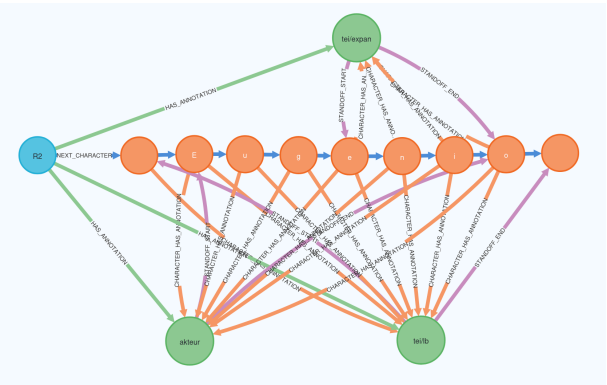
Annotations and text fragments in current digital editions are typically tied to the layout or URL of a single webpage. Furthermore, general URLs are often unreliable due to frequent changes or the unavailability of linked resources (Bleier 2021: 4; Stronks/Boot 2007: 169). As a result, permalinks and persistent identifiers, such as DOIs, have become increasingly important (Bleier 2021: 4). However, permalinks typically point to static content and fail to account for parameterized or dynamic views of texts in digital editions (Stäcker 2020: 9). Additionally, their reliability largely depends on the institution responsible for their maintenance (Bleier 2021: 4). This lack of persistent linking prevents their reuse across multiple

editions or representations of the same text (Koolen/Boot 2020). As a result, fragments are siloed within specific platforms, limiting their scalability and their ability to support broader, cross-edition computational studies.

Early solutions like Canonical Text Services (CTS) URNs aimed to provide persistent, structured references for textual passages, particularly in classical corpora (Tiepmar/Heyer 2017). However, their strictly hierarchical model limits their applicability in networked, overlapping annotation contexts. Similarly, robust hyperlinks as proposed by Phelps and Wilensky (2000) addressed URL instability through lexical signatures but remain tied to surface-level document retrieval and lack structural or semantic depth. These models highlight the foundational need for persistent and content-aware referencing.

Applied Text as Graph (ATAG)

Labeled-property graphs (LPG) allow for the representation of complex relationships and multidimensional aspects. They consist of nodes, edges, and properties, providing a flexible framework for text elements and their connections (Kuczera 2022: 103). The ATAG approach expands this by segmenting text into manageable units within a graph structure. Each character is transformed into a node with a unique identifier, connected sequentially by edges that define their relationships and order (Kuczera 2024).

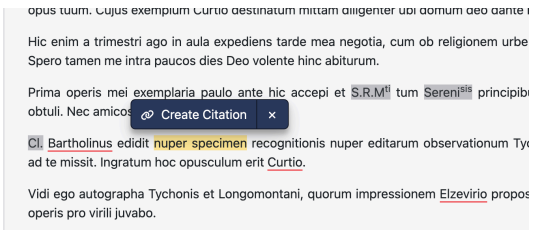


A LPG visualizing the text "Eugenio" with letters as orange nodes connected by edges to green annotation nodes, which are linked to the central text node R2 (Kuczera 2024: 2).

Figure 1 shows a graph fragment for the text "Eugenio" with associated annotations. Orange nodes represent individual characters linked sequentially, while green nodes indicate annotations tied to the text chain, illustrating their relationships and scope. A central blue node integrates all elements, reflecting the character sequence and text properties. Each node is uniquely referenced by a UUID, enabling multiple points of connection in a networked structure.

Explicitly Notated Citations (ENC)

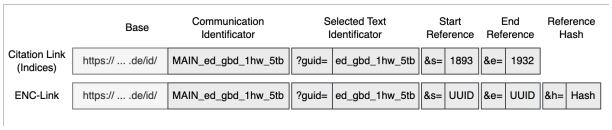
In the digital edition *The Socinian Correspondence*, we introduced dynamic placement of text markers and the creation of citation links, as illustrated in Figure 2. These permanent links enable access to both the broader context and the specifically cited text passage. When the link is used, the relevant page in the edition opens and automatically navigates to the marked section. This method is also applicable to annotations and other objects within the edition, leveraging their persistent identifiers.



Screenshot of the "Citation" feature in the digital edition The Socinian Correspondence, demonstrating the creation of permanent references to specific text passages.

However, a key issue arises as these citation links depend on the start and end indices of the marked text. Changes to the text can lead to broken or inconsistent references, compromising the reliability of the links.

The **Explicitly Notated Citations (ENC)** method is an advancement in citation practices for digital editions (Enns/Kuczera 2025). ENC replaces traditional start- and end-index parameters with unique character identifiers from the ATAG chain. Each citation uses the UUIDs of the first and last characters in the marked section, along with a hash parameter that encodes the entire ATAG chain for integrity verification. If the UUIDs or hash do not match, it indicates a change in the referenced text, and users are notified accordingly. Figure 3 illustrates the structure of an ENC-link compared to the previously mentioned citation link.



Overview of the structure of a citation link using indices compared to an ENC-link, which utilizes unique identifiers instead of indices and includes a hash for verification.

ENC resolves key issues in digital citation by creating stable, dynamic, and granular links. These links point not only to the specific ATAG chain but also to its broader textual context. To ensure persistence, the referenced text and its annotations can be exported in JSON format and versioned using Git, with commit hashes added to the citation link. This allows previous text states to be retrieved and referenced, ensuring that citations remain valid even after changes to the source text. By supporting ENC-links as endpoints for cited texts and annotations in machine-readable formats like JSON, ENC unlocks new possibilities for algorithmic editions. This capability enables automated processing, deeper analysis, and integration with computational tools, advancing the scope of digital scholarly work.

Algorithmic Edition

An algorithmic edition combines ATAG with ENC-links and enables machine-readability and computational engagement with traditional human interaction. Unlike conventional digital editions, which rely heavily on visual interfaces like websites, algorithmic editions also focus on structuring texts and annotations for algorithmic analysis and integration. They prioritize precise access to textual content, annotations, and contextual metadata. Every element – texts, annotations, and their relationships – is uniquely addressable, enabling seamless retrieval and integration into computational workflows.

Using methods such as ENC-links, algorithmic editions allow direct access to specific text segments. These links return metadata like semantic annotations, layout details, and linguistic information (e.g., part-of-speech tags), transforming the edition into a comprehensive data repository for both human and algorithmic use. This granularity supports modern scholarly practices, including detailed citation, intertextual analysis, and exploration of complex textual networks. Such a network-oriented approach views texts as interconnected nodes within broader systems, emphasizing relationships not only between textual elements but also across datasets (Witt 2018).

Notably, an algorithmic edition does not require a visual interface like a website; it can exist as a purely machine-readable structure, enabling integration into diverse digital ecosystems. This flexibility enhances its usability, fostering interdisciplinary collaboration and innovation. While visual presentation remains vital for many scholars, the underlying structure is optimized for machine interaction, ensuring texts and metadata are equally accessible to human readers and algorithms.

In essence, an algorithmic edition is not merely a digital reproduction of a text but a dynamic system designed for comprehensive exploration. Through its structured approach, it facilitates new insights into both textual content and the cultural practices encoded within (Krämer 2023: 7). It redefines digital editions, integrating texts, data, and technology to advance scholarly practice.

Conclusion

Algorithmic editions redefine digital scholarship by integrating ATAG and ENC techniques to create stable, granular access to texts and annotations. ATAG structures texts as networks of interlinked nodes, while ENC ensures reliable citation through unique identifiers and hash-based verification. Together, these approaches facilitate seamless human and machine interaction, enabling interdisciplinary research and data-driven insights. By treating texts as dynamic networks, algorithmic editions enhance accessibility, computational analysis, and scholarly exploration, setting a foundation for the future of digital humanities.

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Retrocomputing as an Integral Part of Digital Humanities Practice?

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Digitality permeates and shapes contemporary everyday life. However, it is often forgotten that our current understanding of digitality is the result of a long developmental process and, therefore, has a far-reaching historical dimension. While the term "digitization" has only recently gained prominence as a political buzzword, the underlying techno-social transformation has been underway for more than half a century. By the 1970s, with the proliferation of game consoles, and the 1980s, with the advent of home computers that easily integrated into existing household infrastructures (e.g., by using televisions as screens and cassette recorders for loading and storing programs), digitization had already reached broader segments of the population.

Retrocomputing shifts the focus to older computer systems that have been overshadowed by the logic of acquisition and disposal inherent in the computer industry and are now traded as vintage items. Regardless of whether it is an iPhone 14 from 2023 or an Apple II from 1977, computers remain computers and, in principle, can perform the same tasks as envisioned by Turing and von Neumann. However, they are subject to a dynamic historical development that cannot be fully understood solely through the empirically observable increase in processing power (Moore's Law). It must also take into account the diversity of interfaces as well as purpose-driven and target-specific configurations. These devices also include the software executed on them, encompassing the programming languages and techniques used. From this perspective, it becomes clear that earlier concepts and visions of digitality were fundamentally different from today's. At the same time, cultural markers and habitual patterns emerged that persist in some cases to this day. Consider, for instance, the character Donkey Kong or the row-column syntax in spreadsheets, both created or established around 1980. Today's digital culture cannot be fully understood or explained without incorporating its earlier forms and structures.

These historical forms of digitality are the focus of the Retro Computing Lab at the Center for Philology and Digitality (ZPD) at the University of Würzburg (fig. 3). There, the study of old computer systems and their associated games and applications is not conducted through emulators but rather on the original devices. By late 2024, about 15 legacy systems (both home computers and game consoles) will be operational there. The setups, including CRT monitors, floppy drives, and dot-matrix printers, not only offer an authentic approach to hardware and software but also enable an understanding of the historical context in which games and applications were developed, forming the multimedia backdrop of current retro games. Furthermore, they allow access to digital legacies such as unpublished games and their source codes (see Games That Weren't or the NZTronix Collection). This provides students from various disciplines, ranging from computer science to German studies to museology, with expanded opportunities to engage with historical forms of digitality and digital heritage. The location at the Center for Philology and Digitality also integrates this into the bridging (study) disciplines of Digital Humanities and Computational Humanities.

Similar collections and projects can be found at various locations. Notable examples include MITH's Vintage Collection (University of Maryland) and the Media Archaeology Lab (University of Colorado). In Europe, there are also numerous collections, though often outside the scope of Digital Humanities, typically implemented as museums and much less frequently as an integral part of an educational institution or, let alone, a DH institute. This raises the question of whether and to what extent retrocomputing is part of the Digital Humanities.

To address this, I propose four aspects for discussion:

Self-Reflection of DH in the Context of Digitality History

A historical perspective shows that the Digital Humanities primarily operate adaptively, adopting new developments in IT and tailoring them for specific applications in the humanities and social sciences. This forward-looking approach results in high dynamism but also leaves behind a trail of obsolete technologies. The Digital Humanities, whose historiography currently begins with the use of punch cards, are thus themselves subject to an ever-changing digitality. It seems only logical, as part of a professional self-reflection, to engage with the evolution of digitality. Even if computers, as mentioned earlier, remain fundamentally computers, the specific manifesta-

tions of interfaces, data structures, and processing methods significantly shape the digital artifacts left to posterity. This also continually confronts the Digital Humanities with models that must operationalize the complex realities of humanities inquiries.

"Computativity" as Part of DH Didactics

The Digital Humanities could also be embedded in a general cultural history of digital media. However, this perspective might obscure a core constant of DH: the challenge of developing solutions for modeling, analysis, and presentation methods with the computational tools and digital infrastructures available at a given time and place. Problems encountered in representing historical or literary models using text or database models in 1975 could arise similarly in 2025 when continuous information must be translated into a system inherently based on discrete units. Computers, after all, remain computers. Engaging with historical digitalities can thus serve as an abstraction process, revealing fundamental questions of the Digital Humanities independently of their specific or current implementation conditions. Therefore, retrocomputing should be considered not only a historical but also a didactic component of the Digital Humanities.

Fig. 1: Coding practices of the 1980s, type-in programs in computer journals.

Legacy Computing as a FAIR Measure

Beyond self-reflective and didactic uses, retrocomputing offers further possibilities. While "born-digital" objects are increasingly produced, much legacy data is lost due to technological progress, as current software may no longer interpret it, or the conditions of its creation are no longer comprehensible. Retrocomputing could reconstruct original setups to make data visible in their original context, potentially in an emulated environment, and migrate data to interoperable formats with proper documentation. This would help ensure that older and future datasets remain accessible within the FAIR cycle (cfr. Rinne 2011; Feichtinger 2023). Given the technological diversity, it is efficient for retrocomputing centers to specialize in specific areas, such as mainframes, home computers, Windows systems, or Apple systems.

Fig. 2: A Schneider CPC 464 (1984) and data recovered from 3" floppy disks.

Digital Heritage Preservation as Part of Institutional Expertise

Preserving earlier digital systems as operational cultural objects will become increasingly important as future generations produce more digital cultural heritage. Instead of migrating data and removing it from its original context, digital objects could be maintained in their original or emulated environments (cfr. Acker 2021; Schiller/Grigar 2019). However, this requires a high degree of specialization and methodological groundwork at the intersection of DH and computer archaeology (cfr. Schröter 2023) to establish practical and scientific standards.

Outlook

The Retro Computing Lab at the University of Würzburg, still in its infancy, aims to address these aspects in the future. How these can be implemented in practice—through research projects, courses, public outreach, and collaboration with other disciplines such as computer science, media studies, cultural studies, and more—will be revealed over time, as these fields increasingly engage with digital data and cultural heritage.

Fig. 3: The Retro Computing Lab at the University of Würzburg.

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The GOLEM Ontology for Narrative and Fiction

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Introduction

Narrative phenomena can be viewed as interconnected systems in which various components influence one another. Understanding the properties of narratives requires analyzing them in relation to each other and within their broader context, rather than in isolation (Pianzola 2018). To explore narrative structures, computational methods offer valuable insights. Computational narratology focuses on studying narratives through the lens of computation and information processing (Mani 2019). It emphasizes the algorithmic processes involved in the creation and interpretation of narratives, allowing narrative structures to be modeled in formal and computable forms. However, the predominance of probabilistic-quantitative methods in digital literary studies is often insufficient for capturing all aspects of literary phenomena (Ciotti 2016). This limitation necessitates a more nuanced approach, such as a mixed human-machine methodology, like ontology modeling, which also provides a level of abstraction that facilitates the comparison of different stories.

An ontology, defined as a formal, explicit specification of a shared conceptualization (Studer et al. 1998), provides the theoretical and technical means for operationalizing these conceptual frameworks. The GOLEM (Graphs and Ontologies for Literary Evolution Models) project aims to model how narratives adapt across languages and cultures by developing an ontology that represents narratives independently of specific domains. To achieve this, the GOLEM ontology for narrative and fiction establishes a framework defining the interrelationships among key narrative elements, such as characters, social relationships, and events. It creates a library of modules containing these components. In alignment with Linked Open Data principles, the GOLEM ontology is developed as an extension of CIDOC-CRM (CRM) and LRMoo (LRM), two ontologies for cultural heritage, while aligning with the foundational ontology DOLCE-Lite-Plus (DLP).

Related Works

Various ontologies for narratives have been developed over the years. Nakasone and Ishizuka introduced the Storytelling Ontology Model (2006), emphasizing event coherence through Rhetorical Structure Theory (RST). Swartjes and Theune's Fabula Model for Emergent Narrative (2006) take a character-based approach to event sequencing. The Archetype Ontology (Damiano & Lieto 2013) explores recurring narrative roles. The Narrative Ontology (NOnt) focuses on events and is integrated with CIDOC-CRM (Bartalesi et al., 2016; Meghini et al., 2021). The Ody Ontology (Khan et al., 2016) applies temporal structures to literary analysis. The Drammar Ontology (Damiano et al., 2019) focuses on character conflicts. The ProppOnto (Peinado & Iaz-Agudo, 2004) examines character roles based on Propp's Morphology of the Folktale. A related enhanced version is the ProppOntology (Pannach et al. 2021), designed to analyze multicultural folktales and their differences in narrative structures. While these models provide valuable insights, their specialized scope underscore the need for a more extended and reusable ontology.

Methodology

The GOLEM ontology utilizes conceptual modeling and modularization. We begin by defining key narratological concepts, such as character, relationship, and event, linking them to essential elements that describe their characteristics and functions. Our classes are designed for interoperability, connecting with existing ontologies like CIDOC CRM, LRMoo, and DOLCE-Lite-Plus. We establish relationships between these elements and construct the ontology using Description Logic (DL) to facilitate computational inference. Finally, the ontology is used to populate a knowledge graph (Pannach et al., 2024a), allowing users

to perform SPARQL queries for efficient data retrieval and analysis.

Dataset

The current dataset used for the GOLEM knowledge graph is based on fanfiction collected from Archive of Our Own (AO3). We selected AO3 for its accessibility, supportive web scraping policy, and comprehensive metadata system (Pannach et al., 2024b). In line with the ontology's objective to encompass a wide range of narratives, we plan to expand our data sources in the future to include stories from other domains and fictional universes, like folklore.

Module Overview

The GOLEM Ontology includes modules for characters, relationships, events, settings, narrative, and inference (see Figure 1). A wiki and the complete documentation of the ontology can be found on the project's GitHub[1] and on the pyLODE-generated web page.[2]

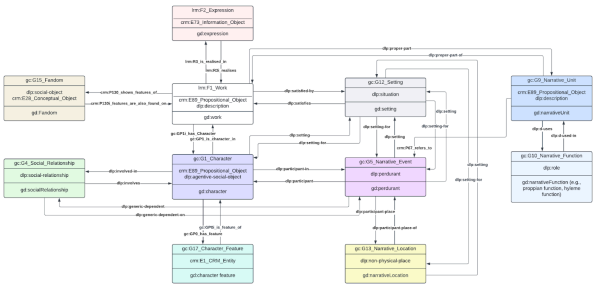

Figure 1. Overview of the GOLEM ontology's main classes and properties.

The character module introduces the class G1_Character, which is a subclass of crm:E89_Propositional_Object and aligns to dlp:agentive-social-object. Characters are viewed as participants within a storyworld, distinct from real-world individuals (Jannidis, 2019). The module also defines the class G0_Character-Stoff (subclass of crm:E89_Propositional_Object), derived from Erzählstoff (narrative material), representing the full range of potential variations for a character across different media (Zgoll et al., 2020). By employing the concept of Character-Stoff, we link all variants of the same character-stoff to a cohesive framework.

While CIDOC CRM provides a class for physical features (E26_Physical_Feature) and quantifiable features (E54_Dimension), it lacks a class for non-physical or qualitative features. Thus, we create G2_Feature, with subclasses G17_Character_Feature and G18_Textual_Feature. G17_Character_Feature includes qualitative character attributes, such as biographical, physical, and psychological traits, while G18_Textual_Feature, is used for aspects like narrative style and point of view (POV).

The relationship module models the social relationships between characters by using dlp:social-relationship. It focuses on the events that shape relationships over time and the roles that characters assume within those relationships (Mika & Gangemi 2016). It regards social relationships as dynamic and evolving rather than fixed (Chaturvedi et al., 2015). Furthermore, this module distinguishes between homogeneous relationships, where all characters share the same role (such as friends), and heterogeneous relationships, where characters have distinct roles. A prime example of the latter is a love triangle, where one character may be linked to two others in different relationships, each with its own distinct role, such as lover and rival. However, the current model does not account for nuances such as the sign (positive or negative) or strength (strong or weak) of these relationships (Mika & Gangemi, 2016).

The event module focuses on the class G5_Narrative_Event, which aligns with dlp:perdurant. This module models two primary types of relations among narrative events (Bartalesi et al. 2017): mereological relations, in which events are parts of larger events, and temporal occurrence relations, which associate events with time intervals to indicate whether one event occurs before, during, or after another.

The setting module models the narrative universe in which a story unfolds. The primary class, G12_Setting, aligns with dlp:situation A situation emerges from a narrative that delineates the setting into spatial, temporal, and social structures that serve the story's needs (DOLCE-Lite-Plus, 2005). Importantly, each setting aligns with the overarching narrative structure of the work, rather than being defined by individual events.

The narrative module introduces key concepts for modeling narrative structures, including G7_Narrative_Sequence (aligned with dlp:course), G9_Narrative_Unit (a subclass of crm:E89_Propositional_Object aligned with dlp:description), G10_Narrative_Function (aligned with dlp:role), G11_Narrative_Role (aligned with dlp:role), and G14_Narrative-Stoff (a subclass of crm:E89_Propositional_Object aligned with dlp:description).

The G9_Narrative_Unit serves as the fundamental building block, encompassing actions, states,

or themes within a story. The G7_Narrative_Sequence organizes these units into structured patterns, primarily through the concepts of fabula and syuzhet. Fabula refers to the chronological sequence of events that outlines "what happens" in a narrative (Abbott 2019). In contrast, syuzhet (often referred to as "plot") focuses on how these events are presented to create effects such as suspense (Kukkonen, 2019; Abbott, 2019). G7_Narrative_Function describes the roles that narrative units play, exemplified by Proppian functions like “Villain causes harm” (Propp 1968). The G11_Narrative_Role specifies the roles of characters, such as "hero" or "villain." Finally, G14_Narrative-Stoff represents the essential materials that underpin a narrative variant (Zgoll et al. 2020).

The inference module addresses the complexities of observational data in narratives by documenting their provenance. Inspired by MiMoTextBase and Wikidata (Schöch et al. 2022), this module aims to capture the diverse interpretations that can arise from a single source. By modeling the relationships between statements, methods, and sources, the module enhances transparency and traceability (Carroll 2015). To achieve this, we utilize the crm:E13_Attribute_Assignment class, which captures the attribution of properties to subjects. The module also incorporates the sources and methods used to generate each statement, allowing one crm:E13_Attribute_Assignment to serve as the source or premise for another.

In addition to its modular structure, the project utilizes the CIDOC CRM class E55_Type to link controlled vocabularies to specific class instances, thereby enhancing interoperability while remaining flexible with respect to various literary and narrative theories. For instance, instances of the class G10_Narrative_Function can take values from the list of "Proppian functions" (crm:E55_Type).

Contribution

The GOLEM ontology plays a crucial role in Digital Humanities by making implicit narrative structures explicit, facilitating computational reasoning and uncovering hidden connections between narrative elements (Ciotti 2016). It provides a structured framework for comparative studies across cultural contexts, enhancing our understanding of cumulative cultural evolution in narratives (Pianzola et al. 2020). Aligned with CIDOC CRM, LRMoo, and DOLCE-Lite-Plus, GOLEM adheres to Linked Open Data (LOD) principles, promoting reusability across research domains. Its modular design allows scholars to apply different modules to specific areas of narrative and fiction studies. As it evolves, GOLEM will encompass a wider range of narratives, enabling analysis of both metadata and textual content, making it valuable for researchers in narrative analysis.

Notes

1. See GitHub repository: https://github.com/GOLEM-lab/golem-ontology
2. See web page: https://ontology.golemlab.eu/

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Postclassical Time Maps: Theory and Interpretation

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This paper fleshes out the theory of "time maps" and demonstrates their interpretive utility. Time maps are the graphs which are produced when a narrative's fabula (i.e. the timeline of events within the story) is plotted against its syuzhet (i.e. the order in which these events are presented to readers/viewers/players). In short, time maps are a digital and data-driven means of visualizing a narrative's temporal relations. My previous work on time maps received the Paul Fortier Prize at the 2019 Digital Humanities Conference in Utrecht for best paper by an emerging scholar. An introduction to the theory of time maps, along with my methodology for producing them, is available in *Cultural Analytics* (Yeager 2023). The present paper discusses three advanced theoretical concepts: narrative velocity; non-linear narrative chronotopes; and ambiguous temporalities. I also offer a comparative analysis of Orson Welles's *Citizen Kane* and Assata Shakur's *Assata: An Autobiography*. Though these two narratives differ in terms of thematics, genre, and medium, they become curious bedfellows when their time maps are juxtaposed.

To start, time maps prompt a re-evaluation of Gérard Genette definition of narratological velocity, which he establishes by comparing " *a length* (that of the text, measured in lines and in pages)" to " *a duration* (that of the story, measured in seconds, minutes, hours, days, months, and years)" (1980: 87-8). He borrows from algebra-based physics, where velocity is the ratio between length and time, to state that narratological velocity is the ratio of syuzhet to fabula. A flaw with Genette's analogy is he is not comparing a length to a time; instead, he is comparing approximations of two different types of time: the character's and the reader's. A consequence of this confusion is that his ratio is upside down, because readers must directly engage with the narrative's syuzhet in order to access its fabula. Instead, it makes more sense to consider syuzhet as the independent variable and fabula as the dependent variable, so that a narrative's velocity is understood in terms of how the fabula fluctuates with regard to the syuzhet. When this understanding is used, the velocity of any given segment of narrative corresponds to the slope of its line on a time map. Figure 1 offers a rudimentary comparison of how this plays out with regard to Genette's five categories of pause, stretch, scene, summary, and ellipsis. In short, the line segments with a steeper slope have a higher velocity (e.g. A sentence-long summary like "she worked all day" has a steeper slope than the full text of *Mrs. Dalloway*, because the latter stretches out the day's events over a few hundred pages).

Figure 1: Comparing the five classical types of narrative velocity.

Pushing this reformulation of narrative velocity means shifting from an algebra-based understanding of velocity to a calculus-based one. Within an algebra based framework, we can only plot line segments on a time map. While line segments can represent many narrative phenomena, they cannot – for example – represent segments of syuzhet where the fabula speeds up or slows down. Consider a film clip which shows characters discussing an impending emergency, then segues into a fast-forwarded sequence of them preparing for this emergency. This storytelling device involves a narratological acceleration – i.e. a shift from one flow of narratological time to another – and we need to use a parabola to accurately represent this shift. My previous work refers to the line segments which get plotted on a time map as "narratological chronotopes;" thinking more broadly requires the incorporation of "non-linear narratological chronotopes."

The final theoretical phenomenon that I consider involves temporal ambiguities which arise when we step outside the realm of "natural" narratives. Consider *The Twilight Zone* episode, "A Kind of Stopwatch." In this story, a character finds a stopwatch which literally stops time for the rest of the world, though it proceeds at a regular pace for him. Time maps are useful for contemplating this kind of structure because there are two potential ways to visualize it: "explicitly adopting some metanarrative frame" or "selecting a privileged viewpoint" (Wittenberg 2013: 7). When using the first strategy, we portray this phenomenon as a Genettean pause, because time for the external world has stopped (Figure 2, left). When using the second strategy, we portray it as a Genettean scene, since time is still flowing for the main character (Figure 2, right). While there is no definitive way to resolve this interpretive impasse, a time map clarifies the problem and prompts a contemplation of the constructed nature of time in narrative.

Figure 2: Two contrasting time maps of “A Kind of Stopwatch.”

In addition to these theoretical contributions, I offer two interpretations which are rooted in time maps. In Genette’s terms, “we can characterize the temporal stance of a narrative only by considering at the same time all the relationships it establishes between its own temporality and that of the story it tells” (1980: 155). Time maps, in other words, are useful interpretive tools because they visualize a narrative’s temporal structure, the embodiment of its philosophical understanding of time.

Consider the time map of Orson’s Welles’s *Citizen Kane* (Figure 3). This graph approximates the shape of a triangle whose sides are the vertical blue dots on the left, the horizontal black line across top, and the multicolored line segments below. These respectively correspond to the opening newsreel, the reporter's interviews, and the events of Kane's life as recounted by the interviewees. The film famously features several narrators, whose spans of syuzhet are demarcated by vertical gray lines, yet these narrators recall Kane's life in almost exactly the order it unfolded, so the triangle’s hypotenuse is nearly linear. As a result, almost every narratological chronotope in Kane lies along the perimeter of this triangle. There is one glaring exception: when Kane is forced to sell controlling interest in his newspaper, the Inquirer , which is shown in red. On a time map, this loss literally lies at the center of the film, and thus shapes the viewer's understanding of Kane's life. The triangle's corners are similarly significant. The film opens in the top left corner when Kane speaks his last word, the bottom left corner depicts him sledding as a child, and the top right corner shows that same sled getting thrown into a furnace. Just as the vertices define a triangle, Rosebud delineates Kane's life. Commentators typically presume that Rosebud represents Kane’s inner child, a wounded and vulnerable core which he attempts to hide behind a facade of “success.” But through the geometric mapping of a time map, Kane's defining feature is indeed his failed ambition and the fixation on his lost childhood is merely a shell of nostalgia that is rooted in cognitive dissonance.

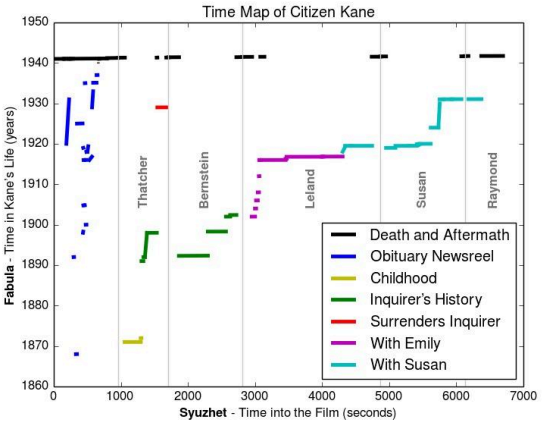

Figure 3: Time map of Citizen Kane .

Interestingly, Assata Shakur uses a very similar temporal structure to Welles in her autobiography (Figure 4), though she deploys this temporality to very different ends. The book opens with a chart which chronicles her trial dates, shown in green. This doesn’t reach all the way back to her childhood, like *Kane*’s newsreel, but it fulfils a similar purpose by giving the reader a preview of what’s to come. Shakur’s first chapter discusses the incident which led to her imprisonment, then her second chapter discusses her early childhood. From there, the chapters alternate between her experiences in prison and her life story, following a “zigzag” pattern which mirrors *Kane*’s alternation between interviews and flashbacks (Montfort 2007: 89). Yet there is a key structural difference between these narratives, which occurs once their internal timelines converge. For *Kane*, the past only catches up to the present posthumously; the end of the character’s life story roughly coincides with the end of the narrative itself. But for Shakur, her storylines instead converge at the moment of her final trial, but this is not the end of her tale, as demonstrated by the tail in the time map’s upper right corner. Instead, Shakur goes on to narrate her experiences of the trial, its resultant conviction, and eventually the details of her life after escaping from prison. The distinction between these two shapes is partially attributable to genre: the narrator of a real-world autobiography must be alive to narrate it, typically in a manner such that “self-distance must end in self-presence as the events of a life catch up with the moment of telling it” (Currie 2007: 61). Yet these theoretical constructs carry over to the narrative’s thematics: *Kane* is the pessimistic story of a rich white man who tries to make himself unknowable to others, whereas *Assata* is the hopeful story of an incarcerated Black woman who tries to imagine a more equitable future – thus the tail.

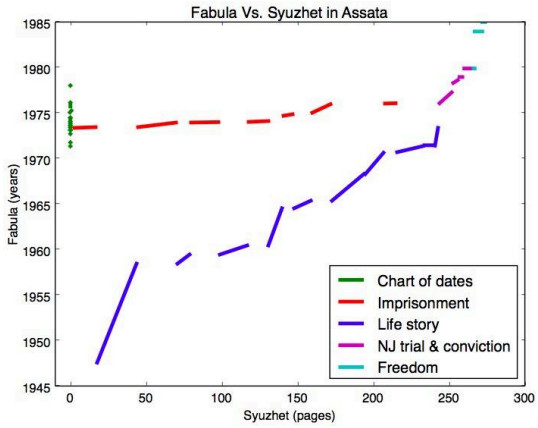

Figure 4: Time map of *Assata: An Autobiography*.

In conclusion, this paper demonstrates two key utilities of time maps. First, they can shed light on abstract narratological structures. Furthermore they offer unique ways to interpret and compare stories, based on their embodiment of temporal relations.

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Revisiting Network Analysis in Drama: Operational Challenges and Methodological Insights

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Network analysis offers powerful tools for exploring character dynamics in dramatic texts, but key methodological choices remain under #specified. This paper evaluates the precision and validity of common segmentation strategies—such as act #scene divisions and speech #based presence—by comparing them against a “gold #standard” derived from fully encoded stage directions.

The use of network analysis for research was popularized by Solomon Marcus in the 1970s (Marcus 1973). As a part of a movement called "Mathematical Drama Analysis," it sought to interpret dramatic texts merely based on their character constellations and the mathematical metrics that could be derived from them. Marcus imagined the process through the lens of a unique spectator, who is only capable of perceiving specific character constellations on stage, which undergo changes through entrances and exits (Marcus 1973: 289–91). These changes can be represented as a binary matrix, which maps the characters onto the scenes they appear in. This matrix, in turn, can be used to represent the play as a co-occurrence network of the characters. Moretti (Moretti 2013) reintroduced this method into literary studies in the early 2010s, which has become one of the central methods within digital drama research (Escobar Varela 2021).

While the method seems straightforward and almost commonplace in Digital Humanities, it relies on some unclear principles. According to Marcus and others, what the binary matrix tracks is the interaction of characters in scenes, with scenes being the segments between entrances and/or exits. In 1988, Draxler had already pointed out the lack of terminological specificity in this approach (Draxler 1988: 11–14). Thus, at least three key components of network analyses are unclear:

- What constitutes an exit or entrance—is a character hiding, sleeping or dead on stage still present?
- In certain ambiguous cases, what constitutes a character—e.g., is the “generic” character identified with the speech prefix "SOLDADO" (Soldier) in the second act of Calderón's *Afectos de odio y amor* identical with any of the soldiers, e.g., "SOLDADO 1.º" (first soldier), in the third act of the same play?
- What does it mean for two characters to have an interaction? As for Marcus, interaction is merely co-occurrence on stage, while for Moretti, "[...] two characters are linked if some words have passed between them: an interaction is a speech act." (Moretti 2013: 214)

These three key components can be found in many plays published by DraCor, one of the most prominent ongoing digital drama research initiatives (Fischer 2019). DraCor segments plays using act and scene divisions and measures "scenic copresence," defined as co-occurrence of characters with assigned speeches within scenes (Trilcke 2022: 568). While this operationalization is practical for large-scale analysis, it relies more on convenience than robust theoretical underpinnings. Scene divisions in dramatic texts are historically contingent, reflecting authorial or editorial decisions, not necessarily marking the span between entrances and/or exits (Pfister 2001: 312-318). Similarly, equating speech prefixes with character presence overlooks nuances like silent or off-stage characters. Scaling up from close reading to distant reading inevitably entails some loss of accuracy. However, understanding and quantifying this "loss" is crucial to ensure that the resulting metrics remain meaningful. To quantify these effects, we select a small corpus of Calderón de la Barca and Christopher Marlowe plays encoded in TEI XML, where every entrance and exit is explicitly marked. This analysis highlights critical trade-offs between operational convenience and methodological rigor. Ultimately, we aim to offer insights into improving network analysis for drama research and refining the balance between scalability and accuracy in digital humanities methodologies.

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A Modest Proposal for Operationalising Dramatic Texts

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Operationalisation in computational drama studies

As evidenced by recent scholarly debate (see e.g. Krautter 2022, Pichler and Reiter 2022), operationalisation remains a core task in any computational approach to literature, and its importance for a robust research pipeline cannot be underestimated. "[B]uilding a bridge [...] from the concepts of literary theory, through a form of quantification, to literary texts" (Moretti, 2013b, 1), operationalisation is a delicate procedure that requires a balancing act between the specificity of the literary text and the algorithmic requirements of the machine. In the context of computational drama analysis – the subfield of computational literary studies that "attempts to model, analyse and interpret dramatic texts using computational methods" (Andresen and Reiter, 2024b, 1) – one could say, with some approximation, that operationalisation practices coincide with the identification and quantification of some crucial dramatic features and the modelling of their relations.

However, defining the 'key components' of a play has always been a complex theoretical problem in itself; the debate began already with Aristotle's comments on the six constituent parts of tragedies (*Poetics* 1450a, 5-10). Currently, many scholars seem to have settled on working with what Kretz (2012) calls the three 'elementary components' of drama: characters, dialogue and plot. Interestingly, a similar division was already introduced by Russian formalist Boris Yarkho, who spoke of three distinctive elements in theatrical texts: 'dialogue' (диалог), 'characters' (персонажи), and 'action' (действие) (Yarkho, 2006, 410).

With some approximation, it can be argued that these high-level components manifest themselves in different textual aspects (content, style, structure...), which in turn can be subjected to different operationalisation practices and explored using different methods. As singled out by Willand et al. (2017), three main approaches seem now prevalent in the field of computational drama analysis: topic modelling of dramatic genres, exploration of network structures, and analysis of character speech.

Measurements and metrics

At a concrete level, operationalisation begins with the choice of the more appropriate features to describe a phenomenon. Most scholars attempting to model dramatic plays have therefore collected readily available metrics (e.g. the number of characters in a play) or developed new measures to capture what they considered key aspects of texts (e.g. the protagonist's features). This second approach should be not considered, however, a goal in itself, but should rather allow to gain a hermeneutical advantage over existing metrics and/or competing non-computational methods. Furthermore, it should be kept in mind that "[m]easures can be taken on any quantifiable aspect of a text, but finding out the significance of that metric for understanding the text, or better, mapping that metric onto a pre-existing critical concept (such as style, plot, or theme), is crucial to making sense of what is being measured" (Algee-Hewitt, 2017, 759).

To revisit some earlier examples of approached for defining metrics in drama, Yarkho's (2006) study on Corneille was based on fifteen features he considered capable of distinguishing comedies from tragedies, while his investigation of speech distribution in five-act tragedies relied on four character-related measures. Interestingly, his quantitatively oriented, but still 'analogical' approach allowed him to create also highly specific features, such as 'saturation' – a compound metric indicating, among other things, 'the percentage of characters acting in their own interest' (Gasparov, 2017, 141-42). Other Russian (neo)formalists followed Yarkho's lead and developed their own metrics. Revzina and Revzin (1971), for example, introduced measures such as 'character connectedness', 'dramatic engagement rank', and 'probabilistic link', which anticipated several later DH efforts; their approach was later reprised by Sapogov (1974). Sperantov (1998), for his part, explored stage directions in Russian verse tragedy through measures such as frequency, average length, presence within verses, lexical diversity, etc.

Similarly, contemporary scholars have kept developing new metrics tailored to their research needs. For example, Trilcke et al. (2017) have taken inspiration from early work by Xanthos et al. (2016) to address the progressive structuring of

plot. To this aim, they computed several 'event-based' (i.e., “identify[ing] or characteris[ing] a particular point in time within a drama”), and 'progression-based' (i.e., “allow[ing] a general characterisation of the transformation of a dramatic network”) measures (Fischer et al., 2017, 2-3). In his quantitative study of English early modern drama, Algee-Hewitt (2017) has instead built on standard network measures of centrality to create two advanced metrics – 'protagonism' and 'mediatedness'.

Vectorisation and its epistemological implications

The previous examples show how scholars often focused on the measurement and comparison of little sets of metrics. However, a deeper comprehension of literary history and cultural evolution could arguably be achieved only by relying on a broader pool of features, covering as many aspects of a play as possible – not only network structures, but also textual size, speech distribution, plot development, etc.

Nonetheless, dealing with multiple features at once is challenging both from a computational and a conceptual point of view. To tackle this problem, one possible approach might be collecting metrics in ordered lists, i.e., in mathematical vectors. So far, the most common use case for vectors in DH research is word embeddings, which aim to capture the meaning of words based on their distributional properties (cf. Firth 1957). [1] Several approaches have been developed to map words to real-valued vectors, ranging from static vector space models and latent semantic analysis to neural networks-based methods, including transformers.

Within drama studies, a similar logic can be followed to construct *play embeddings* – vectors made up of measures that embody different textual aspects of the text. Such embeddings can be easily constructed e.g. by relying on information extracted by the API of the *DraCor* project (Fischer et al. 2019), which calculates a wide range of statistics for each play included in their dramatic corpora (Figure 1). These metrics, describing for instance the properties of character networks or speech distribution among figures, can be collected in vectors which effectively represent the ‘form’ of a pièce, and can be compared to other vectors to measure formal (dis)similarities between dramatic works.

	id	name	yearNormalized	size	averageClustering	density	averagePathLength	averageDegree	diameter
0	greek000026	aeschylus-agamemnon	-458.0	7	0.690476	0.523810	1.476190	3.142857	2
1	greek000004	aeschylus-eumenides	-458.0	7	0.771429	0.619048	1.133333	3.714286	2
2	greek000008	aeschylus-libation-bearers	-458.0	8	0.883333	0.750000	1.250000	5.250000	2
3	greek000022	aeschylus-persians	-472.0	5	0.600000	0.600000	1.400000	2.400000	2
4	greek000011	aeschylus-prometheus-bound	-444.0	7	0.823810	0.476190	1.523810	2.857143	2

Figure 1. A section of the metadata table generated by the DraCor API for the Greek corpus. The numeric values, where relevant, can be collected in vectors to approximately represent the plays’ ‘form’.

While the pipeline for producing play embeddings is relatively straightforward, one might nonetheless question the epistemic validity of the method itself. Compressing texts into a series of numbers that purport to represent some of their most salient aspects, and then comparing these 'bags of numbers' to make claims about literary history, could easily be seen as yet another example of the simultaneously ‘hubristic’ and ‘reductionist’ mindset often ascribed to the digital humanities.

It could be argued, however, that drama vectorisation is just another expression of the 'cultural technique of flattening' Krämer (2023a, 2023b) sees as fundamental not only to DH endeavours, but to humanistic inquiry as a whole. In her vision, the opposition between traditional modes of scholarly interpretation, represented by 'deep' hermeneutics, and data-driven, 'shallow' digital studies is indeed misconstrued, insofar as the humanities have always used 'superficial' representations of cultural objects as their research objects.

Consequently, computational methods seem to differ from traditional ones only in the extension of their 'flattening' procedure, i.e. in the way they transform cultural objects into machine-actionable data. From this perspective, drama vectorisation should not be interpreted as a reductionist attempt to overtly simplify the complexity of theatrical texts, but rather just as a continuation of formalist morphological thinking with modern tools.

Research applications

In general terms, vectorisation could hardly be defined as a novel approach, as many well-established methods in computational literary studies – such as stylometry and topic modelling – rely on transforming texts or words into vectors. This procedure also underpins for what Michael Gavin (2018: 641) calls “vector semantics”, i.e., a “sub-

field of computational linguistics that uses statistical measurements to model the meanings of words". Only recently, however, scholars have begun to explore vectorisation of texts based not on some semantic features, but on their formal characteristics. To wrap up this survey, we briefly review some new promising applications of this research paradigm.

Aside from the pioneering experiments of Solomon Marcus [2] and from the dissertation by Wendell (2021), which used vectorisation only where necessary to replicate some experiments by Yarkho, the first 'fully-fledged' application of vectorisation can be arguably found in Szemes and Vida (2024). Their study, aimed at distinguishing comedies from tragedies in German and Shakespearean drama, created play embeddings based on a set of standard network features and some derived metrics. While their attempt at classification found no major structural genre differences, they were still able to identify some properties predictive of generic alignment, as previously speculated by Ilsemann (1995).

Giovannini and Skorinkin (2024) repurposed this method to study a sample of German and French opera libretti, focusing on exploring their unique formal properties and evolution over time. While their results were limited, confirming well-known challenges in modelling dramatic texts using structural features alone, they did highlight some formal patterns typical of musical drama. Finally, Giovannini (2024) attempted the first comprehensive use of vectorisation to investigate the formal evolution of early modern European drama. By creating comprehensive vectors to represent plays from different traditions and applying several exploratory techniques, he managed to map the evolution of drama over a 150-years timeframe and provide an empirical test of some well-established critical theories.

Notes

1. For applications of word embedding models in DH context see e.g. Schöch (2022); some theoretical considerations in Dobson (2021).
2. For example, he created binary vectors for each character registering their presence or absence from the stage, thus creating a sort of Boolean matrix, and then calculated the difference between vectors (i.e., the difference in stage prominence among characters) via Hamming distance (Marcus 1970, ch. 7).

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Investigating Conceptual Plasticity: On Detecting a Re-Conceptualization of Focalization with Large Language Models

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2020 saw a paradigm shift in Natural Language Processing (NLP) with the release of GPT-3: with the advent of in-context learning – requesting an answer from a model through the mere input of strings – it became possible to use Large Language Models (LLMs) for new, unseen tasks without having to retrain them for these tasks (Brown et al. 2020). In addition to the previously dominant *pre-train-and-fine-tune-paradigm*, the so-called *prompt-and-predict-paradigm* has emerged as an alternative approach (Liu et. al 2023): from now on, it seems possible to adapt LLMs for domain-specific tasks with minimal effort, namely the aforementioned string manipulation or – to use the common expression – through prompt engineering.

In our presentation, we would like to focus on a specific dimension of classification in computational literary studies (CLS) and test whether and – if so – to what extent the *prompt-and-predict-paradigm* does justice to this dimension: domain-specific conceptual engineering. ›Conceptual engineering‹ refers to the practice of describing, evaluating and, if necessary, revising scientific concepts (Cappelen 2018, Chalmers 2020, Koch et al. 2023). In our opinion, this practice is a central characteristic of CLS: the use of digital methods obliges us to operationalize the literary studies concepts used, which presupposes that they are defined in a way that allows this definition to be translated into a rule of measurement (Pichler/Reiter 2022). This repeatedly leads to the intention and extension of technical terms being adapted during operationalization.

Our guiding research question can be formulated as to what extent LLMs are able to adequately realize such re-conceptualization of concepts within the *prompt-and-predict-paradigm*. In other words, we want to examine whether LLMs, when prompted with re-defined concepts from literary studies, are able to ›learn‹ the semantic shift of these concepts and accordingly apply them correctly. We call this ability ›conceptual plasticity‹. In addition, we examined whether a specific prompt-optimizing technique, DSPy (Khattab et al. 2024), can potentially contribute to an improvement in this ability.

For our experiment, we draw on a concept from narrative theory that has repeatedly been the subject of theoretical debate: focalization. Gérard Genette introduced it in his seminal study on narrative in order to replace the previously dominant concept of perspective in a way that allows the distinction between voice and mode of narration. Since its introduction, the term has been repeatedly criticized and redefined in the process (Niederhoff 2014). For our experiments, we take up one of these criticisms and develop a re-definition of the three forms of focalization based on it. The criticism comes from Andreas Kablitz (Kablitz 1988). He takes Genette at his word by stating that there is »focalization [...] wherever what is depicted is related to a process of perception that takes place within the histoire.« In this respect, he rejects Genette's ultimate conflation of questions of perception with questions of information distribution to determine focalization.

Following this criticism by Kablitz, we developed three forms of focalization that are based exclusively on the perceptual processes of the narrative presented: If it is not possible to tie back what is presented in the narrated world to the perception of a single observer, i.e. if what is presented exceeds the perceptual possibilities of a single person, we speak of zero focalization. However, if it is possible or the case that what is presented in the narrated world is bound to the perception of a single observer, we distinguish between internal focalization, when what is presented is bound to the perception of a text-internal figure, and external focalization, when this is not the case, but the presentation could potentially be bound to the perception of a text-internal figure.

Based on this re-definition, we conducted the following experiment:[1] We annotated the first four paragraphs of 26 German *Erzählungen* (mainly novels)[2] for their focalization (*internal*, *external* and *null*).

With the annotated data, we let five different LLMs (GPT-4o, GPT-4.1, o3-mini, Claude Haiku, and Claude Sonnet[3]) in five runs generate an answer for different prompts that requested them to provide a focalization label for a given paragraph. The prompts differed in the following ways: (i)

a prompt with the names of the three labels without further specification of what each label means (Prompt 1), (ii) the labels plus an explanation of what we understand under each label (Prompt 2), which also served as the basis for optimization using DSPy; (iii), the same as prompt 2 with the added information that the given label definitions are re-definitions of the standard focalization interpretation (Prompt 3).

Model	Prompt	F1-Median (Min–Max)	Precision (M ± SD)	Recall (M ± SD)	Accuracy (M ± SD)
GPT-4o	Template_0	0.596 (0.574–0.621)	0.639 ± 0.018	0.583 ± 0.014	0.583 ± 0.014
GPT-4o	Template_1	0.639 (0.619–0.664)	0.648 ± 0.050	0.702 ± 0.016	0.702 ± 0.016
GPT-4o	Template_2	0.635 (0.622–0.654)	0.633 ± 0.052	0.702 ± 0.014	0.702 ± 0.014
GPT-4o	DSPy	0.498	0.586	0.528	0.528
GPT-4.1	Template_0	0.660 (0.652–0.690)	0.735 ± 0.012	0.694 ± 0.013	0.694 ± 0.013
GPT-4.1	Template_1	0.630 (0.606–0.647)	0.571 ± 0.012	0.706 ± 0.018	0.706 ± 0.018
GPT-4.1	Template_2	0.614 (0.614–0.630)	0.562 ± 0.007	0.692 ± 0.008	0.692 ± 0.008
GPT-4.1	DSPy	0.763	0.764	0.774	0.774
o3-mini	Template_0	0.582 (0.576–0.604)	0.663 ± 0.021	0.570 ± 0.011	0.570 ± 0.011
o3-mini	Template_1	0.676 (0.621–0.717)	0.696 ± 0.069	0.689 ± 0.041	0.689 ± 0.041
o3-mini	Template_2	0.656 (0.650–0.700)	0.689 ± 0.022	0.685 ± 0.022	0.685 ± 0.022
o3-mini	DSPy	0.665	0.695	0.651	0.651
Claude 3.5 Haiku	Template_0	0.413 (0.405–0.413)	0.442 ± 0.002	0.442 ± 0.004	0.442 ± 0.004
Claude 3.5 Haiku	Template_1	0.583 (0.583–0.592)	0.603 ± 0.003	0.577 ± 0.004	0.577 ± 0.004
Claude 3.5 Haiku	Template_2	0.581 (0.575–0.582)	0.590 ± 0.002	0.574 ± 0.004	0.574 ± 0.004
Claude 3.5 Haiku	DSPy	0.647	0.653	0.651	0651
Claude 3.7 Sonnet	Template_0	0.672 (0.662–0.682)	0.762 ± 0.007	0.653 ± 0.008	0.653 ± 0.008
Claude 3.7 Sonnet	Template_1	0.767 (0.758–0.784)	0.774 ± 0.009	0.766 ± 0.010	0.766 ± 0.010
Claude 3.7 Sonnet	Template_2	0.753 (0.753–0.764)	0.757 ± 0.005	0.757 ± 0.004	0.757 ± 0.004
Claude 3.7 Sonnet	DSPy	0.753	0.773	0.745	0.745

Figure 1: Evaluation results for all models, given a specific prompt. Shown are weighted F1 score, precision, recall and accuracy.

The evaluation results of all models and the three prompts are given in table 1. Overall, the models achieve moderate (F1 of 0.41) to relatively high (F1 of 0.76) scores, indicating that they are generally able to carry out the task with the given prompts, but still struggle in many places. Prompt optimization using *DSPy* results in improved F1 scores in two out of five cases, performs equally well in two cases, and underperforms in one (GPT-4o), suggesting that while *DSPy* can enhance performance, its effectiveness varies depending on the model and context.

In order to check if the differences in performance between the prompts are above chance level, we conducted McNemar's test and report the results in table 2.[4] We consider a p-value of 0.05 or lower as evidence for the rejection of the null hypothesis.

Model	Total Significant	With dspy	Without dspy
GPT-4o	14	10	4
GPT-4.1	2	2	0
o3-mini	2	0	2
Claude 3.5 Haiku	24	14	10
Claude 3.7 Sonnet	8	1	7

Figure 2: Number of statistically significant differences in pairwise McNemar tests across models, separated by comparisons involving optimized prompts (*DSPy*) and those using standard templates.

Across all five models, McNemar tests revealed varying degrees of sensitivity to prompt differences. For GPT-4o, a total of 14 significant comparisons were found, 10 of which involved DSPy prompts – all of which performed worse – suggesting that this model is particularly sensitive to prompt modifications, though not necessarily in a beneficial way. GPT-4.1 showed only two significant differences, both of which involved *DSPy*, indicating occasional but clear effects of optimization. In contrast, o3-mini exhibited two significant results, none of which involved *DSPy*, pointing to a lack of responsiveness to conceptual explication and prompt optimization. Claude 3.5 Haiku showed the highest sensitivity overall, with 24 significant comparisons – 14 with and 10 without *DSPy* – highlighting strong effects from both optimized and non-optimized prompts. Finally, Claude 3.7 Sonnet showed 8 significant differences, but only one involved *DSPy*, suggesting that this model is primarily affected by differences between standard templates.

Overall, the results suggest that most models – particularly GPT-4.1, o3-mini and Claude 3.7 Sonnet – are relatively unaffected by user-specific prompt optimization and appear to rely on entrenched internal representations of the concepts, which are not easily influenced by manual intervention. GPT-4o and Claude 3.5 Haiku, by contrast, show greater sensitivity to prompt variations, including optimized prompts, indicating a higher degree of conceptual instability. Further investigations should examine whether these findings hold across larger and more diverse datasets, and to what extent different prompt tuning techniques lead to statistically significant improvements in prediction of re-defined concepts.

Notes

1. See https://github.com/AxPic/Reengineering_Focalization_2025 for the experiment's complete code and the prompts used.
2. All of the narratives from which the opening passages were taken are included in the anthology *X* , which brings together canonical narratives of modern German literature: https://www.fischerverlage.de/buch/deutsche-erzaehlungen-9783596903733 .
3. https://platform.openai.com/docs/models/gpt-4.1 , https://platform.openai.com/docs/models/o3-mini , https://openai.com/index/hello-gpt-4o/ , https://www.anthropic.com/claude/haiku , https://www.anthropic.com/news/claude-3-7-sonnet
4. The McNemar test is the standard for determining the statistical significance of two independent classification tests in the NLP, see Card et al. 2020.

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Comparing Human and AI Performance in Visual Storytelling through Creation of Comic Strips: A Case Study

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Abstract

This article presents a case study comparing the capabilities of humans and artificial intelligence (AI) for visual storytelling. We developed detailed instructions to recreate a three-panel Nancy cartoon strip by Ernie Bushmiller and provided them to both humans and AI systems. The human participants were 20-something students with basic artistic training but no experience or knowledge of this comic strip. The AI systems used were popular commercial models trained to draw and paint like artists, though their training sets may not necessarily include Bushmiller's work. Results showed that AI systems excel at mimicking professional art but struggle to create coherent visual stories. In contrast, humans proved highly adept at transforming instructions into meaningful visual narratives.

Introduction and Motivation

Creating visually compelling images has traditionally required significant effort and expertise. Recently, AI-based drawing and painting systems have gained popularity, sparking controversies in the art community, such as a dispute in a cartoon competition (Cumhuriyet, 2024; Darroch, 2017; Kotbas, 2024).

While AI is argued to boost efficiency and create jobs requiring advanced skills (Trivedi et al., 2023), studies show people prefer human-made art for its emotional depth, narrative, and meaning (Bellaiche et al., 2023). AI systems have notable limitations in graphic design (Sindhura and Abdul, 2021). As a solution, hybrid intelligence has been proposed, combining human and AI capabilities to enhance performance, as seen in projects like celestial body classification via crowdsourcing (Kamar et al., 2012) and integrating expert knowledge with AI to improve results (Chang et al., 2017; Bansal et al., 2021).This paper demonstrates that AI struggles to create nuanced visual images for effective storytelling, emphasizing the need for human expertise to craft meaningful visuals.

Context and Motivation

Effective visual storytelling is challenging, even for professionals. Subtle details play a crucial role, as illustrated in Figure 1, which compares two nearly identical cartoon panels on privacy published in IEEE Computer Magazine (Akleman, 2021) 1. A minor change in the eyes transforms mischievous children into surprised ones, shifting the narrative from intentionally reading their sister's diary to an accidental discovery.

Figure 1: An example highlights the role of subtle facial expressions in storytelling. In the left panel, the children appear surprised, likely finding the diary by accident, unaware of their sister's anger as they're engrossed in its contents. In contrast, the right panel shows mischievous children who deliberately searched for the diary, fully aware of their sister's angry gaze.

Effective storytelling relies on creating appropriate human affects, such as emotions, mood, or attachment. For AI, generating such affects is challenging as they involve complex expressive cues

—facial, vocal, or gestural behaviors—learned by humans over decades of interactions (VandenBos, 2007). Subtle cues like body posture or gaze direction can significantly alter the perception of affects (Liu et al., 2012; Akleman et al., 2015; Dede et al., 2024). However, standard research on human affects focuses mainly on broad categories like facial, vocal, and gestural expressions (Ekman and Oster, 1979; Ekman and Keltner, 1997; Ekman, 1999; Russell et al., 2003), struggling to classify subtle and uncommon expressions.

Basis, Rationale and Contributions

Creating human affects is challenging due to the subtle, context-dependent nature of expressive cues, which go beyond simple image and text inputs. Effective storytelling requires considering all sensory inputs within context, as small details like droplets or body posture significantly enhance expression (Akleman and Celik, 2020) (see Figure 2). Subtle cues in storytelling, akin to micro-expressions in emotion recognition, are brief and difficult to synthesize or detect algorithmically (Merghani et al., 2018; Li et al., 2022). While AI tools like facial image generators (Viazovetskyi et al., 2020) produce impressive visuals, minor inconsistencies can disrupt intended meanings (Fan et al., 2022).

Expert artists and animators excel at creating subtle, context-aware cues through practice, though few formally document their methods (Johnston and Thomas, 1981; McCloud and Martin, 1993; Blair, 1995; McCloud, 2006; Eisner, 2008; Celik, 2011). Simplified user studies combined with expert knowledge could improve the understanding and application of these cues, aiding dataset creation (Dede et al., 2024).

Context is equally vital for understanding human interactions and storytelling. Defined in ubiquitous computing as "physical activity, location, and the psychophysiology and affective state of a person" (Bulling and Roggen, 2011), context shapes behavior and norms. Misunderstandings may arise from cultural differences, such as conflicting views on personal space. In human-computer interaction, context-aware systems enhance usability by adapting to specific situations, such as recognizing faster, louder speech in emergencies like house fires, where failure could have critical consequences.

Figure 2: The position and orientation of small props like droplets can alter perceived expressions Droplets near the eyes suggest sadness, on the forehead indicate fatigue, and under the nose imply sickness, even on a neutral face. More examples can be found in

In conclusion, understanding subtle expressions and their context is essential for compelling storytelling. Computational agents, such as AI systems, must analyze and adapt to their operational contexts. This study shows that AI systems still lag behind humans in generating nuanced expressions and contextual understanding necessary for effective storytelling.

Process

Our process for comparing human and AI performance in visual storytelling through the creation of comic strips consists of four steps:

Selection of a Comic Strip: In this stage, the goal is to select well-known comic strips that are recognized and praised by visual storytelling experts for their effectiveness.

Creation of verbal instructions, i.e. prompts: This stage involves describing the chosen comic strip in as much detail as possible using only text. These descriptions, referred to as prompts in the context of artificial intelligence, are essentially verbal instructions for completing a task. Such verbal instructions have long been used by humans to describe tasks.

Creation of Comic Strips by using verbal instructions: In this stage, the same verbal instructions are sent to both humans and various AI systems to produce corresponding comic strips.

Analyzing comic strips in terms of their quality: In this final stage, the quality of the comic strips is analyzed and compared.

This process is general and can be repeated with many different comic strips. However, in this paper, we focus on a single strip, as it is well-known for its quality.

Selection of Comic Strip

To compare the creative capabilities of human and artificial intelligence, we used a prompt based on a famous Ernie Bushmiller cartoon strip (see Figure 3). This strip was chosen because it is extensively analyzed in Paul Karasik and Mark Newgarden's Eisner Award-winning book, How to Read Nancy: The Elements of Comics in Three Easy Panels (Newgarden and Karasik, 1988, 2017). The book, recognized as Best Comics-Related Book in 2018, highlights the strip's artistic significance,

arguing that Nancy, often dismissed as simple, is a cornerstone of comic art.

Karasik and Newgarden dissect this Nancy strip panel by panel, revealing how lines, shapes, timing, and space contribute to its storytelling and humor. They argue that Bushmiller's simplicity masks his precision and mastery. The authors highlight universal comic principles like pacing, visual economy, and text-image integration, showcasing this strip as a model of effective comic art.

They argue that the strip's humor is crafted with mathematical precision, relying on perfect element placement and timing. Bushmiller's methods offer valuable lessons for aspiring cartoonists, demonstrating the storytelling power of visual art and highlighting often-overlooked nuances in comic storytelling.

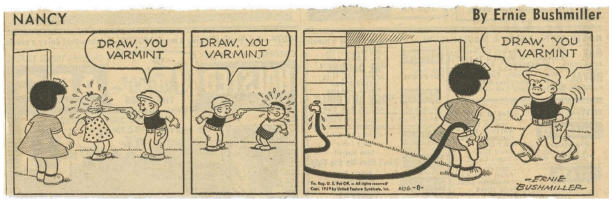

Figure 3: The original "How to Read Nancy" comic strip drawn by Ernie Bushmiller that the prompt is derived from. To the best of our knowledge, this particular strip is in the public domain. While the copyrights for many *Nancy* comic strips are held by Andrews McMeel Syndication, this specific strip does not appear in their published list of copyrighted works. We contacted the syndicate on two separate occasions, and they did not assert copyright ownership over this strip.

We want you to create a three-panel comic strip. Each panel will be positioned side by side. There will be two main characters in the comic strip: "X" and "Y." Instructions for drawing the characters and panels are provided below:

Character Instructions:

- **Character X:** This character is a 10-year-old girl wearing a skirt and a simple T-shirt, with the T-shirt tucked in. She has short, black, curly hair and is slightly overweight.
- **Character Y:** This character is a 10-year-old boy wearing a baseball hat. He wears shorts and a simple T-shirt and carries a water gun, which he uses to wet everyone around him.

Instructions for Panels:

- **Panel 1:** The character X is positioned on the left side of the panel, seen from behind, observing Y as he sprays a girl with his water gun. The character X looks concerned about the situation.
- **Panel 2:** The character Y is in the center of the action, spraying another boy with his water gun. The character X is not visible in this panel.
- **Panel 3:** The character X reappears on the left side, again seen from behind. This time, she has a water hose hidden behind her, ready to retaliate against the character Y. On the right side, The character Y is walking toward the character X with a smirk on his face, oblivious to what's coming.

Table 1: Instructions that are used to create the comic strips.

This strip, celebrated for its wit and depth, is ideal for testing how human or artificial intelligence replicates the mechanics of comics. Its visual economy suggests that transforming illustrations into a cohesive strip reveals the complexity of a specific type of intelligence.

Prompt Creation

Reproducing this particular strip can offer fresh insights into the intelligence required to create the simplest-seeming works. We also expected that it would be possible to evaluate the importance of meticulous craftsmanship provided by Artificial Intelligence in terms of the creation of visual stories.

In the prompt, we specifically converted the names of the characters to X and Y to prevent AI tools from being influenced by character names if they exist in their database and to ensure human artists are not biased by prior knowledge of the "How to Read Nancy" comics. Additionally, some AI tools impose a character limit on the prompts they process. You can see the instructions that are used to create the comic strips in Table 1.

Creation of Comic Strips: AI vs. Humans

After creating the prompt, it was given to three 20-something students with basic artistic training but no experience or knowledge of this comic strip and various AI tools to compare their ability to visualize the instructions. A large set of outputs was generated using AI, but we present only the four best examples here, as shown in Figure 4. The humans, who were unfamiliar with the comic strip, produced three strips, as shown in Figure 5.

(a) An example "comic" created by free model in "3D Animation Style created by Leonardo.Ai(URL-1, 2024).

(b) An example "cartoon" created by custom-trained model "realisticVision" of Stable Baselines.

(c) Two examples created by OpenAI's Dall-E using ChatGPT.

Figure 4: The four "comic strips" created by AI using the prompt.

Analysis of Comic Strips: AI vs. Humans

AI systems demonstrated strengths in generative image creation, especially in character visualization, producing visually accurate images based on prompts. Custom-trained AI models and tools yielded superior visual quality as shown in Figure 4 showing potential when high-end resources are available. However, AI struggled with sequential contexts, such as comic strips, where its limitations overshadowed its strengths.

In contrast, human-generated comics consistently captured the prompt's narrative, showing a bully with a water gun disturbing others and a girl secretly preparing a stronger weapon as shown in Figure 5.These works incorporated subtle nuances that enriched the story, highlighting a key advantage of human intelligence over AI. AI often overlooked these minor yet impactful details essential for enhancing storytelling depth and quality.

(a) The comic strip drawn by Feyza Güreli.

(b) The comic strip drawn by Mücahit Morcol.

(c) The comic strip drawn by Abdullah Raşid Gün.

Figure 5: The three comic strips created by 20-something students with basic artistic training but no experience or knowledge of this comic strip only based on the prompt provided to them.

Conclusion and Future Work

This work highlights a significant distinction between the intelligence demonstrated by AI and humans in creative tasks. AI-generated artworks often exhibit impressive detail and technical precision, yet they lack the fundamental elements required for effective storytelling. In contrast, humans not only understand instructions well but also excel at reproducing the subtle details necessary for visual narratives.

In summary, our findings show that while AI systems are proficient at mimicking professional drawing and painting styles, they fall short in crafting coherent visual stories. Humans, on the other hand, demonstrate exceptional capability in transforming well-defined instructions into compelling and meaningful visual narratives.

There is a need for similar studies, using the same or different comic strips, to track the progress of AI systems in visual storytelling.

Disclaimer

This study was conducted in 2024. A new version of ChatGPT, released in March 2025, began generating significantly improved cartoon strips from the same prompts, although some minor issues with facial expressions remain as shown in Figure 6. This development suggests that much stronger support for visual storytelling can be expected in the near future.

Figure 6: This is an example of the output generated using the prompt in the table with the new version of ChatGPT released in March 2025.

Acknowledgment

We extend our heartfelt gratitude to Abdullah Raşid Gün, Feyza Güreli, and Mücahit Morcol for their invaluable contributions in creating the human sketches based on the given prompts. We are also deeply thankful to Doğukan Kıvrak and Sena Erokyar for their support with paid and custom-trained AI models and for sharing their final outputs.

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The Director's Signature: Stylometry of Theater Choreography via Pose and Action Estimation

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Introduction

The concept of a director's "signature" or unique style is central to *auteur theory*, which posits that a director's personal vision and stylistic choices imprint a distinct identity on their work, making their contribution recognizable regardless of collaborative efforts. The notion is most commonly applied to film studies, but theater studies often holds up directors with distinct visual styles (Robert Wilson, Ivo van Hove, Romeo Castellucci) as examples of "auteurs." Unlike film, where a director has many more options in terms of camera placement, lighting, design, editing, and post-production effects, the live performance director has more limited tools and less control over the work as a whole. The main tool for a theater director is choreography of the actors' bodies and their positioning in space, a distinct phenomenon that theater scholars acknowledge yet rarely analyze. And compared to literary stylometrics, where each word and letter is generally attributable to the intentions of the author, a theater performance comprises many different mediating factors (the script, the stage design, the actors' bodies). A combined distant and narrow viewing of poses applied across multiple productions therefore is more likely to determine a director's consistent choreographic patterns, rather than only considering a single performance.

Related Work

Advances in deep-learning technologies for moving images have benefitted computational theater studies (Bardiot 2021, Escobar Varela 2021) by greatly expanding the data that can be extracted from conventional video recordings, augmenting the traditional "cinemetrics" of shot durations and pixel colors by looking deeper into the frame to identify faces, objects, poses and three-dimensional spatial relations (Chávez Heras 2023). Applying such tools across many recordings of staged performances further enables new "distant viewing" analyses (Arnold / Tilton 2019), including efforts to quantify and compare stage directors' "signature" choreographic styles. The body of scholarship on literary stylometry offers promising questions and methods, although our current effort ventures into non-linguistic realms rarely visited even in previous "beyond authorship" studies of theatrical style that focused primarily on the written word (Craig / Greatley-Hirsch 2017). Our work owes much to efforts to categorize deployments of human poses in visual art history (Kutrzyński / Król 2024), although considering theatrical performances greatly increases the degree of spatio-temporal complexity.

Corpus and Data Preprocessing

We assembled a corpus of 30 full-length theatrical performance recordings, evenly divided among three contemporary "auteur" directors: Bill T. Jones, Romeo Castellucci, and Krzysztof Warlikowski (Appendix 1). We applied an advanced transformer-based pose estimation and tracking tool (Rajasegaran et al. 2022) to each recording, generating both 2D and estimated 3D pose coordinates of figures tracked across consecutive frames. We reduced each estimated pose to the 13-keypoint COCO standard (head, shoulders, hips, arms, legs) for compatibility with downstream tools and also divided each pose track into "movelets" of 1/6 of a second for motion quantification. We further applied an action recognition model that assigns a 60-element Atomic Visual Actions (AVA) embedding vector to each individual pose (Rajasegaran et al. 2023), and also generated a 16-element probabilistic view-invariant "POEM" embedding for each pose (Sun et al. 2020).

The data described above were loaded into an instance of the MIME (Machine Intelligence for Motion Exegesis) framework (github.com/sul-cidr/mime), which provides a web-based video pose timeline exploration interface and is built upon a Postgres database using the pgvector extension to enable fast similarity searching of pose and action vectors and coordinates (Figure 1).

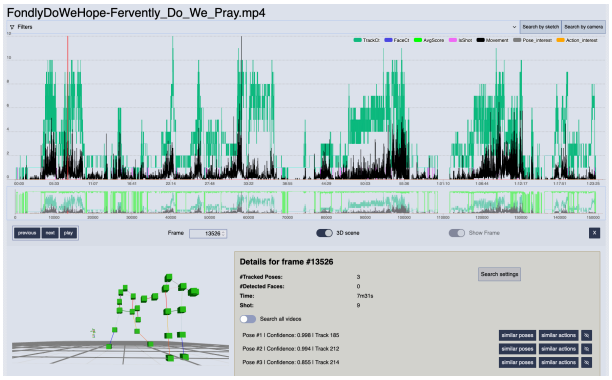

Figure 1. The MIME interface's timeline view, displaying pose and motion data for a full performance, with details and inferred 3D body armatures for a selected frame.

We queried the MIME instance's API to derive a set of pose, action, motion and distance statistics for each performance. Averages (mean and median) and standard deviation of each distribution were calculated per populated frame for the elements of the following:

- POEM pose embedding values
- AVA action recognition embedding values
- 3D "global orientation" representations of each pose (13 x 3 coordinates in meters)
- 3D distances between pose centroids (meters)
- In-place keypoint movement of each tracked pose movelet (meters/second)
- "Sidereal" motion (relative to the background) of each tracked pose movelet (meters/second)

Examining multiple performances per director helps to attenuate the influence of "noise" data from non choreography-related shots of musicians, audience members and post-performance curtain calls (for example), as well as the vision model's struggles with certain camera effects and lighting conditions, and perhaps most significantly, choices of shot angles and framing made by the video directors, who are usually not the same person as the stage director.

Analytical Methods

We framed our primary stylometric inquiry as a performance-to-director classification problem: from the derived pose, action, motion and distance data above, which elements are most effective at predicting the correct stage director for a given performance? We began with "leave one out" (LOO) tests in which we assembled the average pose distance and motion features into a vector for each performance, repeated the process for the 3D "global orientation" pose keypoint coordinates, view-invariant POEM pose vectors and action recognition vectors, and compared these to aggregate vectors that averaged values from all of a director's works except the held-out performance. Simple cosine similarity-based matching of directors to held-out works gave way to machine-learning classification algorithms (a "random forest" combined decision-tree estimator was slightly more accurate than others), which facilitated tests to estimate the relative contribution ("importance") of each element of the feature vectors to successful predictions. Tests were repeated with multiple random seeds and eventually via 10-fold cross-validation to avoid inadvertently cherry-picking results.

	Motion & distance	**Pose embeddings**	**3D global pose keypoints**	**Action embeddings**
LOO cosine similarity	63.33% (19/30)	80% (24/30)	73.33% (22/30)	80% (24/30)
LOO random forest	61% (183/300)	89% (267/300)	66% (198/300)	80.67% (242/300)
10-fold random forest	58.7% stdev: 35%	68% stdev: 35.3%	68.80% stdev: 32.3%	71.60% stdev: 32.1%

Table 1. Accuracies of performance-to-director classifiers by method (rows) and input features (columns). 33% accuracy is expected by chance. Input features are based on mean values of component elements across performances (median values resulted in slightly lower accuracies).

Figure 2. Confusion matrices of directorial classifications of 30 performances using a "leave one out" train/test split across 10 runs with different random seeds.

Figure 3. Feature importances from "random forest" classifiers based on motion and distance (left) and pose embeddings (right).

Results

As Table 1 and Figure 2 (left) indicate, distance and motion statistics aggregated across each performance helped to discern stagings by Bill T. Jones from the others, but the confusion between Castellucci and Warlikowski suggests this may be seizing upon the differences in *genre* between their oeuvres, rather than specific directorial styles: the performances in our corpus directed by Jones are dance works, while the others' tend to be from more static genres, particularly opera. Yet comparing average pose embeddings was largely successful at differentiating between all three directors (see Figure 2, right). Average action embeddings performed similarly (Table 1). This may be due to the tendency of embedding spaces to accentuate semantic differences that are "baked in" during pre-training – which also suggests why comparing the raw global 3D pose data was not as successful, as it does not incorporate such learned biases. The drop-off in accuracy of the embedding-based classifiers during 10-fold cross-validation does however imply that they struggle when the training set is reduced.

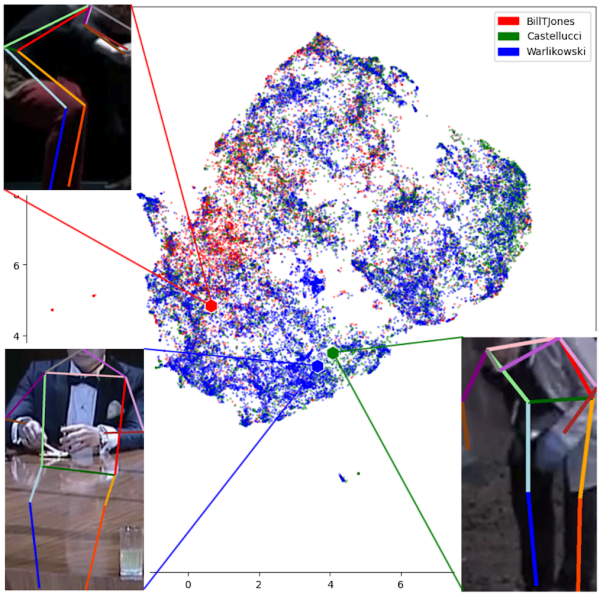

Figure 4. UMAP projection of the POEM embeddings of a 1% sample of poses in the corpus, color-coded by director, showing the locations of the average embedding for each director and sample poses with high similarity to these embeddings.

The superior performance overall of the embedding-based approaches, however, has the drawback that it is more difficult to translate their computational discernment into actual insights regarding the directors' favored poses and action patterns. It is not generally possible to convert an averaged POEM embedding to a single pose, nor is it straightforward to determine precisely what the importance of "feature 1" (Figure 3, right) means for actual poses. Likewise, the AVA action embeddings only map to fairly vague descriptions such as "51%: stand, 38%: watch a person, 33%: carry an object." Yet thanks to the massive vector database available via the MIME platform, it is at least possible to explore poses from the full data set whose embeddings are most similar to a director's averages. Figure 4 suggests this can reveal useful insights, such as the overall greater similarity between Castellucci's and Warlikowski's pose repertoires relative to Jones's, and the tendencies of Jones to employ dramatically contorted poses, Castellucci to favor hunched-over standing postures, and Warlikowski to deploy figures in an active sitting position.

Conclusion

This work demonstrates technologies and methods for addressing the question of directorial style in theater via distant and close viewing of pose choreography. Future iterations will introduce a more diverse set of performances and directors and investigate whether action embeddings can help to untangle the computational challenges inherent to quantification of gesture. Other enhancements will better control for genre and content by comparing different directors' stagings of the same work, and consider scene-level subdivisions of the source performances.

Table 2. Details of the 30 full-length performance recordings analyzed.

Title	Director	Duration	Poses	Tracks	Shots
A Letter to My Nephew	Bill T. Jones	1:09:02	308088	698	182
Analogy/Dora: Tramontane	Bill T. Jones	1:17:12	411046	2136	1794
A Quarrelling Pair	Bill T. Jones	1:17:50	267257	759	729
D-Man in the Waters	Bill T. Jones	0:35:10	141191	855	110
Fondly Do We Hope... Fervently Do We Pray	Bill T. Jones	1:23:55	435160	1001	846
Holzer Duet... Truisms	Bill T. Jones	0:17:45	43206	88	65
Play and Play	Bill T. Jones	1:12:50	462918	2359	230
Secret Pastures	Bill T. Jones	1:33:54	419080	1470	14
Story/Time: The Life of an Idea	Bill T. Jones	1:13:35	402026	1470	869
We Shall Not Be Moved	Bill T. Jones	1:57:30	633482	3244	4111
Das Floß der Medusa (Henze)	Castellucci	1:21:35	114133	724	549
Democracy in America	Castellucci	1:43:05	219148	524	46
Don Giovanni (Mozart)	Castellucci	3:30:01	897915	5155	1009
Go Down, Moses	Castellucci	1:02:40	122960	289	78
Inferno	Castellucci	1:36:16	345711	2837	799
Parsifal (Wagner)	Castellucci	3:58:29	780960	3551	847
Purgatorio	Castellucci	1:13:48	93387	399	392
Requiem (Mozart)	Castellucci	1:38:57	507498	2874	365
Resurrección (Gustav Mahler)	Castellucci	1:40:15	295840	2212	610
The Magic Flute (Mozart)	Castellucci	2:43:15	763697	4385	863
Bluebeard's Castle (Bartók) / La voix humaine (Poulenc)	Warlikowski	2:02:19	278984	1389	736
Die Gezeichneten (Schreker)	Warlikowski	3:48:41	954114	5454	1345
Elektra (Strauss)	Warlikowski	2:06:25	324719	1220	532
Iphigénie en Tauride (Gluck)	Warlikowski	2:11:26	510703	2648	822
Lady Macbeth of Mtsensk (Shostakovich), 1	Warlikowski	1:47:13	412434	3402	774
Lady Macbeth of Mtsensk (Shostakovich), 2	Warlikowski	1:17:36	446255	3619	526
Médée (Cherubini)	Warlikowski	2:24:41	653184	4443	1029
Les Contes d'Hoffmann	Warlikowski	3:08:39	930245	6181	1432

(Offenbach)					
The French	Warlikowski	3:45:47	906390	3410	982
Wozzeck (Berg)	Warlikowski	1:45:20	352588	2766	887

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Stereoscopic Journals: An archive interface entangling diary segments with photo series

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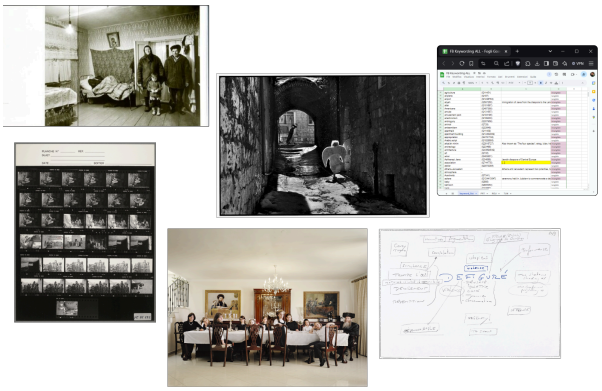

Figure 1: The archive contains various materials, including photographs, diaries, contact sheets, previously curated publications, and metadata.

Introduction

Archives have long been the subject of critical inquiry and, more recently, of experimentation with the new possibilities offered by computation and new media (Baldacci 2016). Today, our memory palaces "have much more permeable walls than their 19th and 20th century predecessors" (Burdick et al. 2016: 49). The ubiquity of the web and digital devices offers unprecedented tools for memory institutions and their visitors to access, interpret, create, and share information, akin to the vision of "living archives" (Dekker 2017).

Within this context, information design emerges as a critical interdisciplinary practice that integrates perspectives from humanities, computing, and design to rethink the form and function of interfaces to our cultural heritage. Foundational principles like Shneiderman's (1996) visual information seeking mantra—"overview first, zoom and filter, then details on demand"—are joined by commitments to serendipity, generosity, and criticality (Dörk et al. 2011; Thudt et al. 2012; Glinka et al. 2015; Whitelaw 2015). These approaches aim to address the diverse needs of different audiences, from researchers to casual visitors, while supporting multiple perspectives and pathways through archival material (Glinka et al. 2017; Windhager et al. 2019).

This research, part of the Granularities of Dispersion and Materiality (GraDiM) project, focuses on the Frédéric Brenner Archive. Brenner, a photographer who spent forty years documenting Jewish diasporic communities worldwide, has amassed a complex collection of over 100,000 photographs, personal diaries, and other materials. These artifacts capture diverse cultural contexts and reflect Brenner's anthropological and artistic approaches (see figure 1).

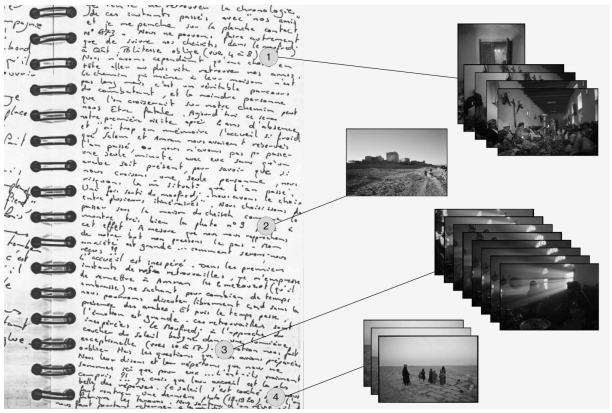

Figure 2: Connections between diaries and photos series in the FB Archive.

Interfacing narratives and data

The archive is rich in narrative potential, particularly through its integration of diaries that provide context, personal reflections, and ethnographic insights complementing the photographic

work. Regarding the relationship between narrative and database, Manovich (2001: 226) suggested the famous definition of "natural enemies", as the two seem to operate in tension. Hayles (2012: 176) reframed this relationship as "natural symbionts," emphasizing their mutual potential when combined thoughtfully. As for the role of texts, Bilansky (2016) pointed out, after Latham (2004), that the rise of the database rather introduced a new form of textuality, both different from the cases of traditional archives and "living literature", and popularized a new way of reading. Arguably, databases can serve as substrates for storytelling, while narratives help structure and humanize data, offering interpretive entry points. Especially in journalistic contexts, storytelling techniques are already used to provide contextual information for data visualizations (Segel / Heer 2010; Hullman et al. 2013).

In this project, Brenner's diaries provide a foundation for exploring the interplay of narrative and archival materials. Personal storytelling, as seen in ethnographic and historical communication, has proven effective in conveying complex themes and fostering emotional connections (Ladeira / Marsden 2014). For photographic collections, researchers have developed methods to extract and reorganize archival content into linear, networked, or screen-like narrative structures (Zeng et al. 2022). These approaches inform the design of an archive interface, Stereoscopic Journals, that combines textual and visual materials in dynamic ways (see figure 2).

Intermediality and design inspiration

Inevitable reference points are drawn from centuries of paper-based textuality, centred around the constraints of print and the format of the book (Drucker 2014). Bibliographic elements like indices, page markers, and captions, rooted in physical necessity, still form central conventions of digital interfaces. The discourse that relates old and new media often emphasizes those with prominent features of intermediality, like cinema (Manovich 2001) and comics (McCloud 1993; Bach et al. 2018), which demonstrate how the interplay of different media can create compelling narratives (Grishakova / Ryan 2011). Another example is the photo-novel: while remaining an underexplored medium, its integration of sequential storytelling, layout, text and photographs can provide a creative model for intermedial interfaces (Baetens / Bleyen 2010).

For the design of Stereoscopic Journals, we draw on these traditions, aiming to present Brenner's archival materials in ways that balance guidance and exploration. A central challenge is navigating the tension between linear narrative flows and open-ended discovery. Inspired by experimental literary techniques like William Burroughs' "cut-up method," which deconstructs and reassembles text to create new meanings, the interface encourages viewers to engage in both structured and serendipitous exploration. It is no surprise that the cut-up technique has been a reference for authors of electronic literature, for its affinity with the combinatorial techniques underlying generative text and non-linear narratives (Hayles 2008).

Designing interfaces that mediate cultural data requires rethinking traditional user interface paradigms. Instead of privileging efficiency and simplicity, "cultural interfaces" (Manovich 2001) can benefit from embracing complexity and multiplicity (Burdick et al. 2016). Contemporary approaches to digital cultural heritage visualizations move beyond search-centric and grid-based displays (Windhager et al. 2019). Instead, they explore multifaceted, interactive information spaces, often incorporating diverse forms of abstraction to encourage a profound exploration and understanding of the represented material. This development aligns with design principles such as serendipity, generosity, and criticality. While serendipity fosters the accidental discovery of connections within the collection (Thudt et al. 2012), generous interfaces aim at revealing its complexity and abundance without overwhelming viewers (Whitelaw 2015). A critical approach makes visitors more sensitive towards absences, biases, and exclusions (Glinka et al. 2015).

Figure 3: The Stereoscopic Journals interface juxtaposes photo series (left) and passages of diaries (right).

Towards a stereoscopic view

Inspired by practices from intermediality and cultural visualization, the design of Stereoscopic Journals integrates two parallel streams: one presenting diary segments and the other displaying photo series (see figure 3). The interface incorporates semantic linkages and scroll-based anchors to synchronize these streams, offering viewers an experience that is both linear and interactive.

Drawing on the prevalence of scroll-based interfaces in data journalism and faceted filtering in collection interfaces, the interface allows the exploration of individual photographs and the perusal of diary episodes, while providing contextual access to larger thematic and relational structures. This dual approach—balancing guided storytelling with open-ended navigation—addresses the diverse needs from casual browsers to dedicated researchers. Key features of the resulting interface are:

- **Dual content streams:** Textual and photographic materials are presented in synchronized vertical streams. The interface allows the reading of diary passages while viewing related photo series side-by-side. This pairing affords a deeper understanding of the contexts behind Brenner's work, and engages visitors through the intermedial storytelling.
- **Semantic linkages:** Semantic connections between diary entries and photographs are encoded as "scroll anchors," acting as keyframes within the narrative. These anchors guide through a curated flow but also serve as access points for deeper exploration.
- **Dynamic folding:** Drawing inspiration from interactive visualization techniques (Brüggemann et al. 2020), the interface incorporates folding mechanisms in both content streams. Viewers can select entities (e.g., people, places, or keywords) to reorganize the interface around thematic relationships rather than temporal progression; as a consequence, the content that is not relevant is folded into minimal slices. Switches between different selections are accompanied by smooth transitions, useful to provide continuity and maintain engagement.
- **Bibliographic affordances:** Borrowing from the traditions of print design, the interface incorporates established conventions such as annotations, cross-references, and visual markers. These elements guide navigation and create a sense of familiarity while leveraging the flexibility of digital media.

The design process involved iterative design, prototyping, and feedback. Preliminary work on Brenner's personal and spontaneous diaries required significant effort to prepare the material for computation. Semantic tagging and database structuring enabled the creation of meaningful links between text and images.

Feedback sessions with collaborators and external evaluators revealed diverse attitudes toward the interface's dual modes of interaction. Some people emphasized the feature of guidance, while others highlighted the serendipity of open exploration. This polarity was viewed as a strength, reflecting the project's goal to cater to multiple preferences. Participants highlighted the playful nature of the interface and its ability to evoke curiosity about the archive's content.

Conclusion and outlook

The Stereoscopic Journals project illustrates how digital tools can transform archives into dynamic spaces. By combining narrativity, interactivity, and intermediality, the interface provides a model for engaging diverse audiences with complex cultural collections. Its design approach underscores the potential of archives not merely as repositories of information but as platforms for storytelling, discovery, and critical engagement.

Looking ahead, the interface offers opportunities for further refinement and application. Expanding its modular framework could allow integration with other archival collections, adapting the system to different types of narrative and data. Additionally, enhancing the platform with user-generated content or collaborative annotation tools could deepen engagement and foster community-driven exploration of cultural heritage.

The Stereoscopic Journals interface exemplifies a contemporary shift in archival practices—one that embraces the richness of ambiguity, multiplicity, and subjectivity. It aligns with broader trends in digital humanities to democratize access to cultural memory while challenging traditional notions of hierarchy and linearity in archival representation. Through this project, the photo archive becomes more than a collection of artifacts—it becomes an active participant in the creation of new meanings and connections.

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‘Flow Filter’: Introducing an upstream exploratory visualisation and filtering of large and detailed datasets.

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The scale and diversity of historical digital data that is becoming available, accelerated by Machine Learning methods of preparation and linkage, and the detail with which it is sometimes modelled, has intensified the challenges and opportunities presented by ‘the computational turn’ in humanities research (Berry 2011). The need for bespoke tools and models for graphical argument construction (Drucker 2020) which support humanistic methods of analysis involving nuance, multi-axial zoom (Butterworth 2023) and multi-perspectival possibilities (Viola 2023) have become more acute.

One particularly significant requirement for ‘generous’ tools and interfaces, (Whitelaw 2015), involves a capacity to work across scale: enabling users to delve into the nuance of data within its local context while also zooming out to position it in broader, global contexts—whether singly or comparatively. However, whilst the authors’ design attention is most often given to those bespoke tools for visualising nuance and complexity, such fine-grain insights are likely to be unworkable at the scale of complete datasets, and so need a prefilter to produce targeted subsets. Having identified this as an essential and generalisable requirement, we subsequently broadened the questions we seek to address. How does someone coming to a richly modelled dataset with some domain knowledge but little familiarity with the schema or properties available for exploration, begin to understand the shape and texture of the data, intuitively and as a productive stage in their research? How can this process be designed as a first stage of increasingly refined exploration, allowing users both to preview possibilities and to reverse and reassess past decisions, while maintaining an overview of the dataset? Without such exploratory tools a researcher may be restricted to well-trodden pathways, driven by prior knowledge and predefined assumptions, with insights into potentially novel connections within a dataset remaining hidden.

The result of these reflections was a set of requirements that re-enforce the need for an exploratory principle of anticipation/preview of comparison/consequence, combined with provisionally and continual iteration for serendipitous discovery, within a large cycle of hypothesis generation and testing. The current range of interfaces available for humanities data is limited and limiting, failing to adequately address these requirements. Existing approaches typically rely on generic search methods, which facet and filter data sets through static menu-based queries and ‘trial-and-error’ workflows.

While this is effective for retrieving specific results, such methods are poorly suited for dynamic exploration, especially by those without prior knowledge of the data. They do not support broader, dynamic exploration of a dataset holistically, in the way necessary to reveal connections, strands and linkages, nor do they allow for interpretative processes (Ciuccarelli / Elli 2019). To meet these challenges, we have adopted a design-based approach of speculative prototyping, or “sandcastle building” (Hindrichs et al. 2019). This method serves as an important mechanism for the construction of argumentation in graphical form (Radzikowska / Rucker 2022) which has informed the design and development of the *Flow Filter*.

The *Flow Filter* is a generalisable exploratory data visualisation component that sits upstream within an environment of multiple different downstream components, each with more specialised analytical application. Drawing on principles from the familiar ‘alluvial’ data visualisation form, the *Flow Filter* functions simultaneously as a visual query builder and a predictive visualisation environment. The interface operates on several levels. It presents a simplified representation of the data structure, for a conceptual overview of the dataset’s organisation. It offers an intuitive interface for serendipitous exploration, encouraging the iterative refinement of research questions. Additionally, its visual rhetorics reveal characteristics and linkages between data attributes in ways that can be shared and communicated across research contexts.

The *Flow Filter* enables the user to construct "sluice gates" through which data is filtered, with each gate assigned a data attribute, with associated properties that can be selected by the user. This visual logic integrates two aspects of query

logic: “OR” logic applies when adding gates, while “AND” logic governs the selection of properties within gates. ‘Flows’ connect these gates, dynamically illustrating relationships and distributions between data attributes. The width of each flow line is proportional to the volume of data passing through it, visually mapping the shape of the data: its structure, densities, and interconnections.

The interface is interactive and responsive to user intuitions and curiosity. The selection of ‘flows’ and ‘bars’ can be modified dynamically, with queries always updatable, enabling researchers not only to *view* determined search results but also *anticipate* potential pathways and connections within the dataset. By previewing the breadth or narrowness of downstream ‘flows’, users are guided toward areas of interest for more detailed analysis: whether their interest foregrounds more representative or more anomalous data, or the relationship between the two. As familiarity with the dataset grows, even the non-expert rapidly begins to be able to derive useful insights and formulate questions.

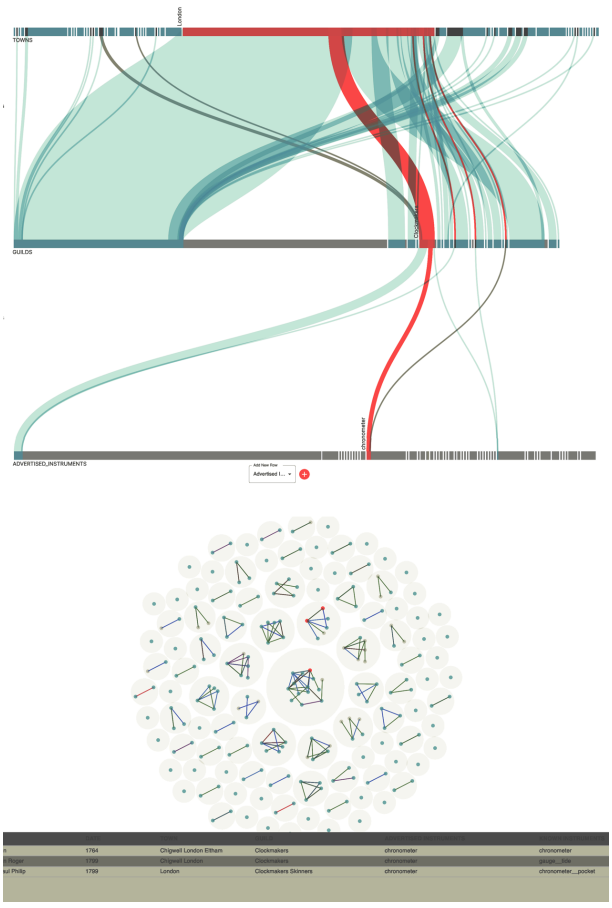

Figure 1. A screenshot of a ‘Flow Filter’, illustrating connections between scientific instrument makers based in London, the Guilds they belong to and the instruments they advertised making.

The paper will demonstrate both the use and intrinsic effectiveness of the *Flow Filter* as a discrete interface, and as the generalisable upstream component that is applied to three datasets which can integrated with their more specific downstream interface components, to enable an expanded scope for iterated exploration.

The datasets used for the demonstration are: the ‘social clusters’ of residents distributed between 3,500 households in the Saltaire ideal town in the later nineteenth century; the social and business networks of scientific instrument maker communities in Britain from the mid-eighteenth to mid-nineteenth centuries (around 12,000 individuals); and the individuals involved, in varied roles, in the flights from Britain into wartime France in support of resistance networks.

Particular attention will be given to the first of these, for exemplary purposes, around the refinement of broad record sets into meaningful subsets. It will demonstrate how households can be probed and filtered based on attributes such as census year, household size, the ‘identity’ of households (as annotated following network science analysis), gender, place of birth and occupational types and subtypes (as variously categorised by different classificatory systems). It will show how *flowlines* can reveal trends like household size distribution over time or the relationship between gender and occupational roles, prior to more sophisticated analysis of selected subsets.

By advancing interactive visualisation methods, *Flow Filter* contributes to digital humanities research, advancing an intuitive and engaged approach to data exploration which has broad applications in an era of large-scale data and knowledge graph technologies.

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The irreductionist hermeneutics of the Grounded AI Map

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The *Grounded AI Map* is a giant "walkable visualization" (Rodighiero, 2018) of how AI and algorithms are involved in science, designed by analyzing 2 million scientific papers retrieved from Scopus (Fig. 1 & 2). The map consists of a 100 square meters floor mat that participants can explore physically by walking on its surface and interact through a smartphone app. It shows one dot for each paper and four thousand labels of thematic clusters. The visualization has been exhibited in multiple public venues in 2025, including the Danish Technical Museum (Fig. 3), and has been adapted for audiences in both Danish and English. The mobile website includes three modalities of information: text, audio, and visual data.

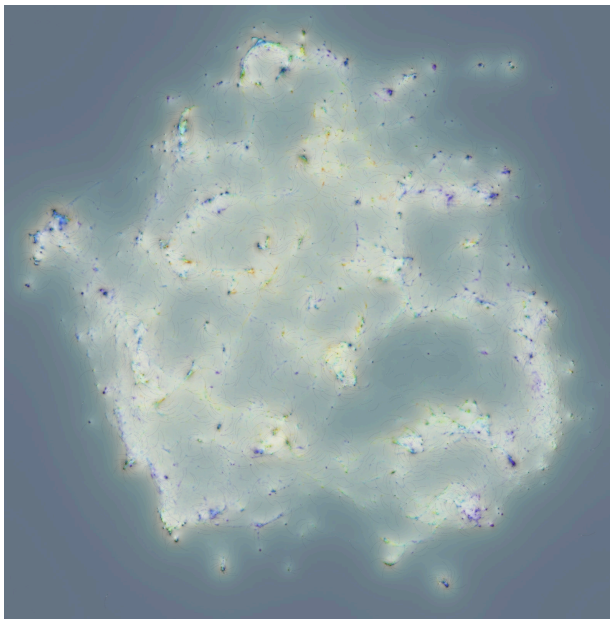

Figure 1: The *Grounded AI Map* visualizes 2 million scientific articles extracted from Scopus.

Figure 2: Close-up of the map: each dot is a scientific article, annotations indicate specialties.

Figure 3: Exhibition at the Danish Technical Museum.

Exhibition at DH2025

We propose to exhibit the installation at the DH conference, producing a smaller scale version if necessary. Our primary intention is to engage the public by providing a first-hand aesthetic experience featuring a distant reading that allows the visitors to move throughout different scales of abstraction of the 2 million scientific papers mapped.

Method

We harvested a corpus of approximately two million scientific articles from the *Scopus* database, using the following query in the title or abstract: *algorithm*, AI, "artificial intelligence", or "machine learning"*. The query is deliberately limited, yielding an incomplete yet broad representation of the AI discourse in science.

We computed connections between articles using text analysis and natural language processing techniques. We associated each abstract to a vector using an embedding model trained on scientific literature (Cohan et al., 2020). Then we extracted a network of semantic similarity from the vectors using the k-nearest neighbors approach, where nodes are documents and edges are semantic similarities.

We computed a placement of the documents using a force-driven layout algorithm (Jacomy et al., 2014), allowing clusters of similar documents to emerge. We rendered the map by only displaying the nodes, as dots, color indicating the publication year, from red (old), to yellow, and to blue (recent).

We extracted the densest clusters in the embedding space to identify over 4000 scientific specialties. Using a large language model and a specific prompt, we summarized how AI and algorithms are involved in each specialty, and generated a title. We validated this method qualitatively and quantitatively. We used the titles to annotate the map and make it readable without aggregating the data points, avoiding a reductionist approach.

Design philosophy

We focused on a hermeneutic engagement with the corpus through different forms of interaction. Although our method involves computing semantic distances with an AI embedding and retrieving clusters of similar documents, the initial data is never reduced to those clusters but instead annotated with them to minimize the loss of information.

We call this approach “irreductionist”, following the notion that phenomena and documents are neither reducible nor irreducible in essence (Latour, 1984). We do not reject all reductions (and our map uses some), but we reject more broadly that meaning is necessarily based on reduction. We conceived our map as a mediator between a corpus of 2 million papers involving AI and algorithms, and the viewers, specifically aimed to provide a reversible translation by making use of reduced affordances (annotations) and the non-reduced source (the dots).

We argue that despite the reduction of two millions documents into four thousands annotations, the presence of the dots themselves allows engaging with features that would otherwise get lost in reduction. For instance, the clusters aren’t entirely separate entities as suggested by the algorithm; they often overlap, stretch, collide, and generate numerous unique patterns that can be directly observed on the map. We attempted to implement “graphical means to produce interpretative work using visual argument structures such as contradiction, ambiguity, parallax, and point of view that are fundamentally hermeneutic in character” (Drucker, 2018).

We accompanied the map with a website and smartphone app offering support for engaging in dialogue with it (Fig. 4). It features the “slanted bots”, AI agents with distinct personalities: one optimistic and progress-oriented, the other sceptical and critical of AI. Commenting on each cluster when navigating the app, they contribute to keeping the meaning of the map open to interpretation.

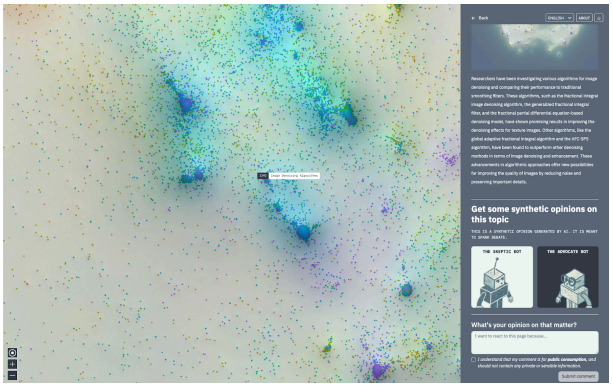

Figure 4: Online app allowing to navigate the map and interact with the “slanted bots”.

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Simple visualisation techniques for simplified Humanities. A survey of Digital Humanities projects

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Introduction

Generous interfaces (Whitelaw 2015) and storytelling tools (Renda et al. 2023) are argued to support interpretation in Cultural Heritage, yet humanistic knowledge cannot be easily embedded into visualisation systems (Drucker 2011; 2014). Scholars advocate for visualisations exposing interpretative aspects (Drucker 2011; 2014; Windhager et al. 2019; Trocchianesi / Bollini 2023; Elli et al. 2021; Viola 2023), encouraging slow visualisation (Elli et al. 2021; Hinrichs et al. 2019; Bradley et al. 2016) and epistemic potential (Münster / Terras 2020; Schwan et al. 2019).

Domain-agnostic taxonomies and classifications describe interaction techniques (Shneiderman 1996; Yi et al. 2007; Dimara / Perin 2020; Brehmer / Munzner 2013) and narrative design spaces (Segel / Heer 2010; McKenna et al. 2017; Zhao / Elmqvist 2023). Though uncertainty visualisation taxonomies (Kinkeldey et al. 2014) informed DH research (Panagiotidou et al. 2023), surveys reveal limited attempts to visualise uncertainty, scarce narrativity, and use of multiple visualisations (Khulusi et al. 2020; Windhager et al. 2019; Panagiotidou et al. 2023). However, studies remain narrow-scoped, disregarding approaches to humanities enquiry. This work addresses uncertainty in DH projects and introduces novel classificatory aspects for tailored visualisations.

Methods

We analysed 186 web-based DH projects collected from the literature (Windhager et al. 2019) and follow-up projects (33 projects), DH associations' websites (AIUCD[1], EADH[2]) and research centres (104 projects), and the Digital Humanities Awards website[3] (49 projects). Selected entities include accessible web projects without access barriers that significantly rely on visualisations.

Our iterative methodology, inspired by (Segel / Heer 2010), developed a classification schema through direct observation and interaction with the visualisation systems, minimising bias from discrepancies between documented features and actual implementation. The framework adapts previous models, covering: 1) narrativity and narrative strategies (McKenna et al. 2017); 2) humanities domains (Terras at al. 2016: 138; Schreibman et al. 2004); 3) visualisation techniques (Windhager et al. 2019); 4) interaction techniques (Yi et al. 2007; Dimara / Perin 2020; Brehmer / Munzner 2013); 5) uncertainty visualisation methods (Kinkeldey et al. 2014; Panagiotidou et al. 2023; Drucker 2011); and 6) critical adaptations, identifying tailored visualisations that embrace humanities-specific complexity, promoting slow looking inquiry.

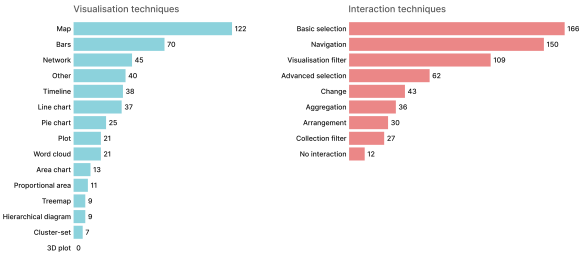
Three research questions guided our analysis:

- **RQ1.** How do DH projects employ information visualisation, interaction, and narrative techniques?
- **RQ2.** How do DH projects address humanities visualisation needs?
- **RQ3.** How do DH projects differ from projects in other humanities-related contexts?

Results and discussion

The dataset (Battisti 2024) is analysed in a dedicated Jupyter Notebook (Battisti 2025).

Our findings confirm a naive reuse of visualisation techniques. DH projects use more maps and fewer networks compared to previous studies (Windhager et al. 2019; Khulusi et al. 2020; Panagiotidou et al. 2023), likely due to network readability issues (Koponen / Hildén 2019). Interactive strategies are limited to easy-to-deploy options, with some non-interactive visualisations present (Figure 1). Specific visualisation and interaction patterns were identified using the Apriori algorithm[4], while a small positive correlation exists between the number of visual and interaction techniques (Spearman coefficient=0.394; p-value=2.54e-08).

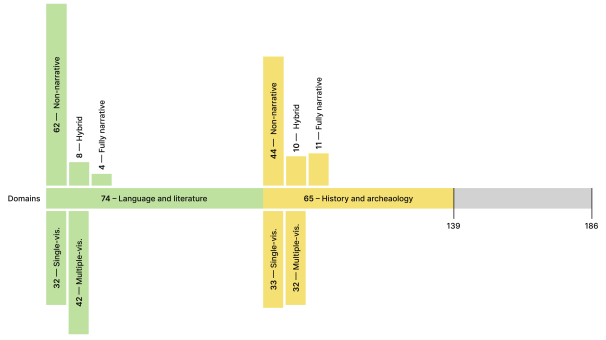


Frequency of visualisation and interaction techniques.

Multiple visualisations are less used than in Cultural Heritage collections (Windhager et al. 2019). Narrative strategies remain underutilised despite growing advocacy (Gershon / Page 2001; Renda et al. 2023). When present, narratives are linear and author-driven, often using tools like StoryMapJS[5] and TimelineJS[6] with minimal customisation (Figure 2).

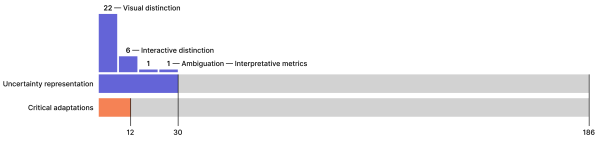
Distribution of narratives and decomposition in narrative progression (top) and type (bottom).

“Language and literature” and “history and archaeology” are the most represented domains, each characterised by narrative strategies and visualisations usage (Figure 3).



Narrativity (top) and multi-visualisation approaches (bottom) in "language and literature" and "history and archaeology" domains.

Few DH projects critically adapt visual solutions, with uncertainty visualisation remaining underrepresented (Khulusi et al. 2020; Panagiotidou et al. 2023) (Figure 4). Despite arguments about researchers' goals and skills determining approaches to uncertainty (Ma / Xiao 2021), our results confirm a persistent theory-practice gap (Windhager et al. 2019; Panagiotou et al. 2023), reflecting broader DH trends favouring simplistic visualisations repurposing over interpretative, critical approaches rooted in humanistic epistemologies (Allington et al. 2016; Windhager et al. 2019; Viola 2023).



Distribution of critical adaptations and uncertainty representation methods.

Acknowledgements

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What Happens When "Hacking" Becomes Easy? Teaching Python in 2025

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In the "big tent" days of the Digital Humanities (DH), debates over "hack versus yack" and whether DH scholars should always be "builders" underscored the field's tension between technical expertise and interpretive scholarship. Since 2022, however, the emergence of AI coding tools is reshaping the stakes of this conversation. Writing Python scripts no longer requires years of practice; now, a well-crafted prompt can produce code in moments. While programming appears more and more attainable for non-technical users, especially those who lack the necessary resources for learning how to code, other considerations may be falling to the wayside. As many have pointed out, when a tool automates a task (e.g., data cleaning), users may not notice its assumptions or limitations, leading to overly simplistic interpretations of complex phenomena.

This shift poses urgent questions about how Python education and research projects should adapt to this new landscape. What happens when hacking becomes easy? Does the streamlining of technical processes diminish the intellectual labor of coding, or does it free scholars to focus on the interpretive and conceptual dimensions of their work? If traditional coding education involved mastering challenging skills and overcoming high barriers to entry, what new forms of rigor emerge when these barriers are lowered?

This panel explores the implications of this paradigm shift from the perspective of four seasoned Python instructors, whose roles span graduate student, professor, librarian, and industry professional. While some of the perspectives emphasize the importance of foundational coding skills for cultural and feminist critique, others advocate for flexibility in evolving with technological changes that lower barriers to creating and sharing work. The panel calls for a reconsideration of what it means to teach, learn, and critique code in an age of AI assistance, ensuring that DH scholarship remains both accessible and intellectually rigorous. Together, these perspectives highlight the evolving role of Python—and programming more broadly—in DH.

PAPERS

Dr. Filipa Calado is an Assistant Professor at the Pratt Institute School of Information. Her presentation explores how AI technology can be re-purposed not to automate or streamline tasks, but to engage directly with underlying biases that drive these tools. She argues that close attention to the mechanisms of coding and the assumptions that circulate within computational processes can illuminate how bias operates in social and discursive contexts more broadly.

Filipa deploys AI to interrogate its own biases in her research project, which uses Large Language Models (LLMs) to study discourses of transphobia in the US. For this project, she trains an LLM with examples of transphobic text, culled from current “anti-trans” legislative bills that are proliferating across the US, with the purpose of examining the bias and discrimination that result in its output. Each step of data gathering and model development opens the logics and assumptions behind machine learning processes to critical analysis which can lead to surprising realizations. For example, prediction algorithms, which turn semantic meaning in language into numerical probabilities, what Filipa calls a “regularization” or “approximation” of language, reveals unexpected commonalities between polarized political perspectives, surfacing shared investments across transphobic and gender-affirming positions. In this context, AI tools are deliberately deployed not for efficiency or productivity, but as a means of turning them back on themselves, offering new objects and rich opportunities for critical analysis.

Dr. Patrick Smyth is a staff engineer and training specialist at a top five venture-backed security startup. Patrick argues for a renewed focus on the fundamentals of coding as essential to cultural criticism. He asserts that a deep understanding of programming principles allows DH scholars to interrogate the assumptions embedded in computa-

tional tools, from the algorithms driving data analysis to the biases encoded in AI systems.

Patrick argues that large language models are fundamentally backward-looking— lowering the cost to produce work based on previous inputs moves us toward a conservative, stultifying, and ultimately mediocre technocultural environment. Further, the development of technologies such as LLMs have fractured the interfaces, such as public APIs, that enabled productive tinkering and remixing, as organizations such as the New York Times or Reddit are racing to close the gates on an outflow of information to AI companies. While code can be quickly written, that easily-written code rarely matters, as algorithmic and platform incumbents tighten the screws on communities of practice. Ultimately, AI saps energy from communities of builders, creating a new set of intermediaries and impediments between communities of practice and meaningful change. Patrick believes that, while AI is here to stay, programmers must reinforce concepts such as platform autonomy, interoperability, cultivated illegibility, and disintermediation.

Stephen Zweibel is Digital Scholarship Librarian at the CUNY Graduate Center. He supports digital project creation by GC researchers across disciplines , helps preserve those projects , and supports faculty and students with their data-based research and data management needs .

Co-programming with AI assistants reframes how humanities students learn to code: the emphasis moves from rote syntax to directing, critiquing, and refining machine-generated code. By shifting from local Python setups to browser-runnable JavaScript apps—and deploying via GitHub Pages or Observable—students bypass installation hurdles and immediately share live projects. First, learners build foundational JS fluency (control flow, DOM manipulation, data structures); then they integrate LLM tools (e.g., CLine, Roo Code) to scaffold web applications. Each generated snippet becomes an object of critical analysis—students interrogate design choices, expose biases, and iterate accordingly. This two-stage model preserves scholarly rigor while democratizing web-based DH practice: LLM-enhanced co-programming extends, rather than replaces, humanistic inquiry.

Rafael Davis Portela is a Ph.D. Candidate in History at the CUNY Graduate Center, researching the History of Computing and of the Computer Industry.

Rafael points out that what and how we teach is always changing, and this is not something that AI is forcing us to deal with for the first time, although LLMs might force us to change faster. While we are in the middle of a transition, and don't know exactly how DH and computing at large will look like in the future, we know we have to adapt. It is a great time to experiment and to force ourselves to try different things. At the same time, we also need to be cautious and recognize that technological changes are often overstated in what they will do in the short term. We should avoid rushed conclusions and the urge to throw away pedagogical practices we have developed over the past decade. Situating his comments in a larger context of the History of Computing, Rafael concludes that Python is, after all, still a dominant language and understanding code continues to be incredibly important in this post-LLM world.